

Tracing Data Through Analytical General Equilibrium[†]

Minji Bang[‡]

Hanbaek Lee[§]

University of Cambridge

May 2026

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Abstract

We develop a heterogeneous-household framework with closed-form individual and aggregate dynamics jointly capturing growth, business cycles, and distributional change. Three moments (aggregate capital and two wealth-sorting covariances) suffice for the equilibrium, which is invariant to household forecasts and transversality. The model is exactly invertible at both macro and micro levels: it reads observed data as an equilibrium path and recovers the primitives, prices, and distributional states consistent with it. In U.S. aggregates (1950–2019), wealth concentration emerges from TFP-driven return-wealth sorting, and feeding the recovered shocks forward forecasts output and hours more accurately than a reduced-form time-series model. At the household level, inverting PSID and SCF microdata yields the joint distribution of individual parameters, which reproduces observed life-cycle and cross-sectional patterns (a hump-shaped age profile of productivity, sharp wealth sorting) and agrees across the two surveys. Counterfactuals on these households show that persistent capital-tax changes produce decades-long distributional drift, with the gains from growth accruing disproportionately to the wealthiest.

Keywords: Heterogeneous agents, closed-form aggregation, wealth inequality, business cycles, forecast invariance.

JEL codes: D31, E21, E32.

[†]We are grateful for helpful comments and discussions from Konstantinos Angelopoulos, Timo Boppart, Vasco Carvalho, Minsu Chang, Joel Flynn, Eric French, Elisa Giannone, Tomohiro Hirano, Benjamin Larin, Rasmus Lentz, Claudia Macaluso, Marta Morazzoni, Florian Trouvain, Tommaso Porzio, Vincenzo Quadrini, Federico Rossi, and Johannes Wieland, as well as the seminar participants at Kyoto Institute of Economic Research and the University of Glasgow. All errors are our own.

[‡]Email: mb2503@cam.ac.uk

[§]Email: hl610@cam.ac.uk

1 Introduction

“... all agree that it began with tracing an outline round a man’s shadow.”

— Pliny the Elder, *Natural History*, Book XXXV.15 (trans. Rackham, Loeb)

The postwar U.S. economy has shown sustained growth, recurring business cycles, and a steady rise in wealth inequality, which are studied apart, yet they are driven by common forces and feed back on one another. Productivity growth raises returns; persistent high returns make already-wealthy households wealthier; and a more concentrated wealth distribution changes how the economy absorbs shocks. The forces behind this joint behavior are not observed. We see only the outline they cast in output, capital, hours, and the cross-section of households without the true figure behind it.

This paper reconstructs that figure behind the observations in general equilibrium, at two scales (the aggregate economy and the individual household), and without first detrending or filtering the data. The obstacle is that standard heterogeneous-agent models are available only as numerical solutions that map parameters into simulated data, so confronting them with data requires stationary, pre-filtered series and an outer estimation loop. We instead solve the model in closed form. The analytical policies and the exact aggregate law of motion make the mapping from structural drivers to data explicit, and explicit enough to invert: the structural drivers are read directly off the raw data, period by period and household by household.

The framework rests on a preference structure that combines well-known ingredients in the literature (log utility and a Stone–Geary subsistence level, augmented by a wealth-dependent time endowment) under which the household problem admits closed-form policies that aggregate exactly. Although households face idiosyncratic productivity and return risk, the otherwise infinite-dimensional wealth distribution collapses to three summary statistics: aggregate capital and two covariances measuring the alignment of wealth with productivity and portfolio returns, sufficient for all contemporaneous equilibrium allocations and prices. We call the resulting object a *moment-recursive competitive equilibrium* (MRCE). Because the household problem is solved in closed form and aggregates analytically into an exact aggregate law of motion, and because the equilibrium is invariant to households’ forecasts about aggregate dynamics, the framework admits an exact data inversion: it constructs the sequence of structural objects under which the observed path itself is a general-equilibrium allocation. Growth and business cycles are thereby captured jointly along a single non-stationary path, with no detrending or stationarity assumption, and heterogeneity is resolved as finely as the data permit, down to the level of the individual observation.

The paper makes three contributions. First, we develop a utility form which allows

closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic productivity and return risk. The utility features a Stone–Geary subsistence level that scales with effective labor productivity, and an effective time endowment that increases in wealth. The individual optimal path yields a unique MRCE in which three moments (aggregate capital and two covariances measuring the alignment of wealth with productivity and with portfolio returns) are sufficient for all contemporaneous equilibrium allocations and prices, and next-period aggregate capital is an exact closed-form function of those three moments. The saving rate is increasing and concave in wealth (Dynan et al., 2004). The framework nests several benchmarks: shutting down return heterogeneity yields Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). With return risk alone, the wealth distribution drops out as in Angeletos (2007); adding productivity heterogeneity restores its role through exactly two covariances.

Second, the equilibrium is invariant to household forecasts about aggregate dynamics. Under log utility, income and substitution effects cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over forecasts, making the MRCE a forecast-invariant recursive competitive equilibrium (FRCE). The FRCE accommodates non-stationary paths without detrending, which is essential for studying how TFP *growth*, not just TFP levels, drives inequality. Conventional approaches either compare stationary equilibria at different TFP levels or require expectations over the entire future price sequence. The FRCE sidesteps both: any TFP path can be fed into the forward simulation to obtain valid equilibrium dynamics.

Third, the closed-form structure enables an *exact general equilibrium inversion* that operates at two scales. At the aggregate scale, given U.S. output, capital, and hours (1950–2019), the aggregate equations trace the model-implied path of TFP, saving behavior, labor efficiency, prices, and distributional moments under which the observed series are themselves the equilibrium allocation, replacing the reduced-form residual accounting of Chari et al. (2007) with a structural path inside a heterogeneous-agent general equilibrium. At the household scale, taking the recovered aggregate prices as given, the same closed-form policies invert PSID and SCF micro data into each household’s productivity, return, and patience parameters wave by wave. This is the most novel use of the framework: rather than recovering a few aggregate shocks, the inversion recovers the entire joint *distribution* of individual structural parameters, observation by observation, with no distributional assumption imposed: an object that forward-solving methods calibrate or approximate but do not measure. Two independent surveys agree on the recovered distribution: the median β is within 2 pp across PSID and SCF. Because the recovered objects are structural parameters in a model with explicit distributional dynamics, they support counterfactual decompositions that isolate

each shock’s role in inequality growth, deliver conditional forecasts of output and labor that beat a reduced-form time-series benchmark, and pass a Lucas-critique-style test that no household’s patience parameter shifts around four major postwar capital-tax reforms.

The inversion yields several substantive findings. A *sorting accelerator* channel links TFP growth to wealth concentration: higher TFP raises returns, which compound through near-unit-root return-wealth sorting and widen the wealth distribution via self-reinforcing feedback (Piketty, 2014). At the baseline calibration, the channel’s unconditional amplification of a one-period TFP shock is modest. It becomes quantitatively decisive under persistent policy shifts: the 1981 and 2001 capital-tax reforms produce wealth-distribution divergences that widen for decades, because the near-unit-root sorting mechanism prevents mean reversion (Section 5.1.4). The traced primitive sequence reveals a two-dimensional business cycle. TFP is nearly orthogonal to both labor productivity and patience (year-on-year correlations of 0.142 and 0.105, respectively). Labor productivity and patience, by contrast, are strongly negatively correlated (-0.858): labor-efficiency declines coincide with rising patience, an anti-cyclical pairing consistent with a precautionary-saving channel. The structural classification attributes individual postwar recessions to different shocks, with this labor-efficiency/patience pairing appearing as one recognizable but not universal pattern.

Related literature Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) showed numerically that mean wealth nearly suffices for aggregate forecasting; the MRCE identifies the omitted moments and gives exact conditions under which the approximation succeeds. The result is closely related to Den Haan and Rendahl (2010), who show that cross-sectional averages of individual-policy basis functions are natural aggregate states. Here those averages reduce to economically interpretable sorting covariances. Computational approaches to high-dimensional heterogeneous-agent dynamics include Reiter (2009), Winberry (2018), Ahn et al. (2018), and the sequence-space Jacobian method of Auclert et al. (2021). The contribution here is complementary: it gives exact nonlinear dynamics for a restricted class rather than linearized dynamics for a broad class.

The paper also builds on exact aggregation under log preferences. Brock and Mirman (1972) solve the representative-agent growth model, and Angeletos (2007) obtains aggregation with uninsured return risk. This paper shows how adding labor-productivity heterogeneity restores a role for the wealth distribution through two covariances. Bilbiie (2020) and Ravn and Sterk (2021) develop parsimonious heterogeneous-agent models with analytical aggregate dynamics; the MRCE uses a continuum of agents with rich return heterogeneity. Alvarez and Lippi (2022) develop analytical impulse responses using eigenvalue methods in

menu-cost models; the moment structure here plays an analogous role for heterogeneous households.

Krueger et al. (2025) and Ando et al. (2025) achieve closed-form transition paths in a limited-commitment economy where the zero-income state reduces each type’s problem to cake-eating under log utility. The MRCE takes a different tractability route from the same log-utility foundation: rather than eliminating labor income, it folds labor income into the effective return through a Stone–Geary wealth-leisure complementarity, preserving continuous idiosyncratic distributions, return heterogeneity, and endogenous labor supply. The trade-off is that the limited-commitment structure delivers a closed-form transition for K alone, while the MRCE’s covariance dynamics require agent-based simulation. The two approaches are complementary: they are dual entry points into analytical heterogeneous-agent macro, each covering ground the other cannot.

The forecast-invariance result is related to block-recursive equilibria (Menzio and Shi, 2010) and cognitive discounting (Gabaix, 2020), but arises endogenously from the income-substitution offset under log utility. This is the same property that Krueger et al. (2025) and Ando et al. (2025) exploit when noting that current policies are independent of future interest rates. Forecast invariance is thus a shared engine across multiple tractable heterogeneous-agent structures, not a feature unique to any one market arrangement.

Forecast invariance also speaks to a recent challenge to the field. Moll (2025) argues that rational expectations are the weak link in heterogeneous-agent macroeconomics: households are required to forecast the equilibrium law of motion of an infinite-dimensional distribution, an assumption that is neither computationally innocuous nor empirically supported, and survey evidence shows large, systematic belief biases that move aggregates (Bhandari et al., 2025). The MRCE is largely insulated from this critique. Under log utility the income and substitution effects of a change in the *forecast* of future prices cancel, so optimal policies are independent of that forecast: the infinite-dimensional expectational fixed point disappears, and with it the requirement that households solve it correctly. Beliefs do not vanish from the model, however. The same cancellation that removes the forecast leaves patience and expectation indistinguishable from current behavior, since a household that saves more because it is patient and one that saves more because it is pessimistic about future returns look identical. Belief content is therefore carried by the recovered discount factor: the individual-specific, time-varying β absorbs shifts in a household’s effective valuation of the future. This makes β a *semi-structural* object, structural up to an expectational component the inversion leaves bundled, whose separation is the subject of Section 5.3 and a natural target for future work.

The household inversion connects to a parallel program in macro-labor that recovers

latent worker and firm types from micro data. [Bonhomme et al. \(2019\)](#) classify latent types from matched employer–employee panels using finite mixtures; [Hagedorn et al. \(2017\)](#) establish nonparametric identification of sorting models from wages and labor-market transitions; and [Lentz et al. \(2023\)](#) and [Hong et al. \(2026\)](#) recover worker-type dynamics and sorting via hidden-Markov classification. Our micro inversion is methodologically distinct (it reads structural parameters off closed-form equilibrium policies rather than estimating a latent-type mixture), but it shares the same goal of recovering individual-level structural heterogeneity from observed cross-sections, and our [Table 7](#) decomposition into permanent, life-cycle, and transitory components is a direct counterpart to the type-versus-dynamics decompositions in that literature.

The state-dependent amplification results contribute to the nonlinear business cycle literature: [Storesletten et al. \(2004\)](#) document countercyclical cross-sectional risk, [Bloom et al. \(2018\)](#) study uncertainty-driven state dependence, and [Berger and Vavra \(2015\)](#) analyze wealth heterogeneity in consumption dynamics. The asset pricing results relate to idiosyncratic risk and equilibrium returns ([Constantinides and Duffie, 1996](#); [Schmidt, 2016](#)); growing evidence on heterogeneous portfolio returns ([Fagereng et al., 2020](#); [Bach et al., 2020](#)) motivates the model’s return channel.

Roadmap The remainder of the paper proceeds as follows. [Section 2](#) sets up the model. [Section 3](#) derives the MRCE, establishes forecast invariance, and draws out its theoretical implications. [Section 4](#) implements the inversion at two scales: aggregate (U.S. output, capital, and hours, 1950–2019) and household level (PSID and SCF), and validates both. [Section 5](#) traces the macro and micro implications: aggregate dynamics, the sorting accelerator, conditional forecasting, tax-reform counterfactuals, household parameter distributions, and a Lucas-critique-style policy-invariance test. [Section 6](#) concludes. Proofs, extensions, and additional applications are collected in the Online Appendix.

2 Model

This section sets up the environment and defines the equilibrium; the analytical results that exploit it follow in [Section 3](#). The economy is otherwise a standard incomplete-markets, heterogeneous-household model. A continuum of infinitely lived households face idiosyncratic labor-productivity and portfolio-return risk and choose consumption, saving, and labor supply each period. A representative firm hires capital and labor in competitive markets, and a government taxes factor income and rebates a lump-sum transfer.

The only departure from the textbook setup is in preferences, where utility is logarithmic

with a subsistence floor that scales with the household’s own earnings and a time endowment that rises with wealth. The production side, factor pricing, and market clearing are entirely conventional, and together with the household problem they close the economy as a recursive competitive equilibrium (Section 2.7).

The purpose of this structure is tractability without giving up heterogeneity. The preference departure is precisely what lets the household problem be solved in closed form, with consumption, saving, and labor policies linear in the individual state. The otherwise infinite-dimensional wealth distribution then aggregates into a few moments (Section 3), and that closure is what later allows the equilibrium to be inverted on data rather than solved forward. We keep every other ingredient deliberately standard, so the results trace to the preference structure rather than to auxiliary modeling choices.

2.1 Preferences

Household preferences take a Stone–Geary form over consumption and leisure with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \quad (1)$$

where $g(x; X) = \kappa w(X)(1 - \tau) Z_L z$.

Here c is consumption, n is labor supply, a is wealth, z is idiosyncratic labor productivity, and $\theta > 0$ weights leisure. The household state $x = [a, z, e]$ and the aggregate state X are specified in Section 2.2, and the wage $w(X)$, return $r(X)$, and aggregate labor productivity Z_L are determined in equilibrium (Sections 2.4–2.5). In the usual way, utility is logarithmic: consumption enters net of a subsistence requirement $g(x; X)$, and leisure is the gap between a time endowment and hours worked.

Two ingredients depart from the textbook form, and each earns its place by making the household problem aggregate. First, the subsistence requirement $g(x; X)$ is tied to the household’s own after-tax earnings rather than fixed in consumption units. This scales the floor with work-related expenses and keeps the consumption policy homogeneous along a growing path, so the framework accommodates trend growth without detrending; a fixed floor would instead inject a non-homothetic level that breaks exact aggregation as the economy grows. Second, leisure is measured against a *wealth-dependent* time endowment $\kappa + \xi a$, with base endowment $\kappa \in (0, 1)$ and $\xi > 0$ capturing wealth buying usable time through time-saving services (Aguilar and Hurst, 2007). This wealth-leisure complementarity links the cross-sectional wealth distribution to aggregate labor supply, and is why a distributional statistic enters the aggregate labor and capital equations rather than dropping out; $\xi = 0$

severs the link.

The log curvature, though standard, also does structural work: it fixes the saving share out of effective wealth at a constant independent of the return, because the income and substitution effects of a return change cancel exactly. This is what underlies the forecast invariance of Section 3.4 and the linear wealth policy that aggregates. The three preference parameters (κ, θ, ξ) are calibrated jointly with the rest of the model in Section 4.2.

2.2 States and aggregate environment

Each household's state is $x = [a, z, e, \beta]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous, with $\mathbb{E}[z] = 1$), a return-scaling factor e (exogenous, with $\mathbb{E}[e] = 1$), and the household's discount factor β (a structural patience parameter, possibly varying both across households and over time). All four display substantial cross-sectional variation (Kuhn et al., 2020a; Storesletten et al., 2004; Fagereng et al., 2020). The productivity factor z scales effective labor, so a household supplies $Z_L z$ efficiency units per hour worked, where Z_L is the aggregate labor-productivity index described below. The return factor e scales the gross return on wealth around the economy-wide average. Both are normalized to unit mean, so all cross-sectional dispersion in earnings ability and portfolio returns is carried by the deviations $z - 1$ and $e - 1$; their transition laws appear in Section 2.6.

The aggregate state is $X = [\Phi, A, Z_L, \bar{\beta}, B, C]$: the joint cross-sectional distribution Φ over (a, z, e, β) , aggregate TFP A , aggregate labor productivity Z_L , the capital-weighted aggregate discount factor $\bar{\beta} \equiv \mathbb{E}_\Phi[\beta_i a_i R_i] / \mathbb{E}_\Phi[a_i R_i]$, return dispersion B , and productivity dispersion C . The aggregate $\bar{\beta}$ is what enters the closed-form aggregate equations of Section 3 and what the macro inversion of Section 4.2 recovers period by period; the underlying individual β_i distribution is recovered wave by wave from PSID and SCF in Section 4.3. For notational simplicity, when there is no ambiguity we write β for both the individual state and its capital-weighted aggregate; the context resolves which is intended. In the quantitative application (Section 4), A , Z_L , and $\bar{\beta}$ vary at the aggregate level; B and C are held at steady state. The framework accommodates time-variation in all five (e.g., widening return dispersion during financial crises or shifting patience during demand-driven episodes). The idiosyncratic β is a household-specific primitive whose cross-sectional distribution is part of Φ ; its transition, together with those of z , e , and wealth, is collected in Section 2.6.

The aggregate labor-productivity index Z_L captures aggregate labor-market conditions: utilization intensity, matching quality, and workforce composition. It enters production only through the efficiency-weighted labor aggregate $L \equiv Z_L \int z n(x; X) d\Phi$ introduced in Section 2.4; it does not appear as a separate factor multiplying $L^{1-\alpha}$. This asymmetry

distinguishes Z_L from neutral TFP A : a shift in Z_L rescales effective hours one-for-one but leaves the production technology itself unchanged (see Section 4 and Online Appendix B for the structural interpretation). At steady state $Z_L = 1$.

2.3 Choices and constraints

The recursive formulation of the household's problem is:

$$v(x; X) = \max_{c, a', n} u(c, n, a, z; X) + \beta \mathbb{E}v(x'; X') \quad (2)$$

$$\text{s.t.} \quad c + a' = w(X)(1 - \tau)Z_L z n + (1 + r(X)(1 - \tau)e)a \quad (3)$$

$$\Phi' = \Gamma_\Phi(X) \quad (4)$$

$$a' \geq 0, \quad n \geq 0 \quad (5)$$

The household's problem does not include the transfer T : households are unaware of its existence and optimize based solely on factor income. The actual law of motion for wealth is $a'_{\text{actual}} = a'_{\text{chosen}} + T(X)$, where T is added by the government after the household's decision (see Section 2.5). The wage $w(X)$ is per hour of labor and $r(X)$ is the aggregate return on capital. The factor e scales returns: $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average. The product $Z_L z$ determines the household's effective hourly wage; Z_L captures aggregate labor market conditions while z captures individual heterogeneity. The discount factor $\beta > 0$ may vary across households and over time, capturing shifts in patience due to financial conditions, uncertainty, or other demand-side forces. We do not impose $\beta < 1$: the household problem is well-posed for any $\beta > 0$ as long as effective wealth is positive, and what a stationary wealth distribution requires is a condition on the aggregate multiplicative root, $\mathbb{E}[\log \beta R_i] < 0$ (Corollary 1), not that every individual β lie below one. An individual $\beta \geq 1$ simply describes a household that is accumulating, as wealth-poor households empirically do; it does not violate the model. The household faces a borrowing constraint $a' \geq 0$, which is shown to be non-binding directly from the household's optimization (Corollary 1 below).

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK_{eff}^\alpha L^{1-\alpha}, \quad (6)$$

where the inputs are *efficiency-weighted* factor aggregates:

$$K_{eff} \equiv \int e a d\Phi = K + \text{cov}(e, a), \quad L \equiv Z_L \int z n(x; X) d\Phi, \quad (7)$$

with $K = \int a d\Phi$ denoting physical (wealth) capital and $\alpha \in (0, 1)$. The return factor e_i scales the efficiency of household i 's capital in production (e.g., reflecting portfolio quality or entrepreneurial ability), just as z_i scales labor productivity. Factor prices satisfy:

$$r(X) = \alpha A K_{eff}^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1 - \alpha) A K_{eff}^\alpha L^{-\alpha}, \quad (8)$$

where δ is the depreciation rate. Competitive factor markets pay $(r + \delta)$ per efficiency unit of capital and w per efficiency unit of labor. Household i supplies $e_i a_i$ efficiency units of capital and $Z_L z_i n_i$ efficiency units of labor, earning $(r + \delta)e_i a_i - \delta a_i$ and $w Z_L z_i n_i$ respectively. Aggregating: total capital income $(r + \delta)K_{eff} = \alpha Y$ and total labor income $wL = (1 - \alpha)Y$, exhausting output. Because factor prices depend on K_{eff}/L , distributional forces (both labor supply shifts and return-wealth sorting) feed back into prices.

2.5 Government policy

The government levies proportional income taxes at rate τ_K on capital income and τ_L on labor income, and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot [\tau_K(r + \delta)K_{eff} + \tau_L wL], \quad (9)$$

When $\tau_K = \tau_L = \tau$, this reduces to $T = \eta \tau Y$ (since $(r + \delta)K_{eff} + wL = Y$). After-tax prices are $\tilde{r} = r(1 - \tau_K)$ and $\tilde{w} = w(1 - \tau_L)$.¹ Remaining revenue finances government consumption (not in utility or production). The baseline sets $\tau_K = \tau_L = 0.30$; the Online Appendix considers time-varying rates.

The transfer plays a dual role. First, it is a fiscal policy channel: tax revenue is partially redistributed, with η controlling the degree of redistribution. Higher η compresses the wealth distribution and thins the Pareto tail. Second, it provides the additive intercept in the

¹Government revenue in formula (9) is computed on gross capital income $(r + \delta)K_{eff}$ — consistent with U.S. corporate taxation of book profits net of depreciation — while the household's after-tax return $\tilde{r} = r(1 - \tau_K)$ is defined on the net return r . The implied wedge between household tax payments and aggregate revenue is $\tau_K \delta K_{eff}$, equal to about 1.5 pp of capital income at baseline ($\tau_K = 0.30$, $\delta = 0.07$). This wedge is absorbed in the lump-sum transfer T and does not affect any household first-order condition or aggregation result; we retain the formulation because it preserves the linear separability $a' = \beta R_i a + T$ on which Theorem 1 depends.

Kesten process governing wealth dynamics, which is necessary for a stationary distribution with a Pareto tail to exist (Kesten, 1973). Without $T > 0$, the purely multiplicative system $a' = \beta R_i a$ collapses to the degenerate distribution at zero and admits no non-degenerate stationary distribution.

Timing The transfer arrives after the household's consumption-saving decision: households solve their problem without knowledge of T , and the government adds T to wealth at the end of the period. This timing is essential for preserving the analytical tractability. If households anticipated T , they would incorporate its present value into effective wealth, breaking the $v = A \log a + B$ value function conjecture and the multiplicative-additive separation $a' = \beta R_i a + T$ that enables closed-form aggregation. The assumption is quantitatively innocuous: at the baseline calibration, the per-period transfer is 0.1% of mean wealth, and the present value of the entire transfer stream is approximately 2.6% of mean wealth. Incorporating this present value into effective wealth would shift saving shares by a correspondingly small amount.

2.6 Exogenous individual state variables

The two exogenous individual states follow stationary AR(1) processes in levels with unit mean:

$$z' - 1 = \rho_z (z - 1) + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, \sigma_z^2), \quad (10)$$

$$e' - 1 = \rho_e (e - 1) + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0, \sigma_e^2), \quad (11)$$

where σ_z, σ_e are the per-period innovation standard deviations and $\varepsilon'_z, \varepsilon'_e$ are mutually independent, so the unconditional variances are $\sigma_z^2/(1 - \rho_z^2)$ and $\sigma_e^2/(1 - \rho_e^2)$. Linearity in levels is essential: it is what makes the cross-sectional covariances $\text{cov}(z, a)$ and $\text{cov}(e, a)$ sufficient statistics for contemporaneous equilibrium allocations (Section 3). In the quantitative analysis each process is discretized using the Tauchen (1986) method.

The endogenous individual state, wealth, evolves through the budget constraint. Given the household's chosen saving and the subsequent government transfer, next-period wealth is $a' = a'_{\text{chosen}} + T(X)$; Section 3 derives the closed-form expression for a' as a function of the current state. The discount factor β is fixed at the household level.

The aggregate state evolves through the distribution, $\Phi' = \Gamma_\Phi(X)$, where Γ_Φ applies the household saving policy and the exogenous idiosyncratic processes to the current cross-section. Constructing Γ_Φ is the central difficulty taken up in Section 3.

2.7 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given Γ_Φ .
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.
4. Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ .

The difficulty in this definition is concentrated in the aggregate law of motion Γ_Φ . Household policies depend on the aggregate state only through current prices (w, r) and the aggregate indices $(A, Z_L, \bar{\beta})$, but prices are set by market clearing, which depends on the entire cross-sectional distribution Φ through the efficiency-weighted factor aggregates $K_{eff} = \int e a d\Phi$ and $L = Z_L \int z n d\Phi$. The distribution is an infinite-dimensional object, and not merely because it has many points: what matters for the aggregates is the joint distribution of wealth with productivity and with returns, so the relevant state encodes the entire pattern of cross-sectional sorting, not just the marginal distribution of wealth. A full RCE solution therefore requires tracking Φ and forecasting its evolution, the standard computational obstacle in heterogeneous-agent economies (Krusell and Smith, 1998): to forecast prices, agents must forecast the future distribution, and closing that forecast generally demands either a finite-moment approximation of Φ or a conjectured aggregate law verified at a fixed point. The next section shows that this economy escapes the obstacle exactly rather than approximately.

3 Moment-Recursive Competitive Equilibrium

In a heterogeneous-household economy, equilibrium prices depend in general on the full cross-sectional distribution of wealth, an infinite-dimensional object that must be tracked and forecast. We now show that the model of Section 2 avoids this exactly. Because the household problem is logarithmic with a subsistence floor and a wealth-dependent time endowment, the optimal policies are linear in the individual states, and a distribution that enters market clearing through linear policies is summarized by its low-order moments. The cross-sectional distribution enters only through three: aggregate capital K and the two

sorting covariances $\text{cov}(z, a)$ and $\text{cov}(e, a)$. These are sufficient statistics for all contemporaneous equilibrium allocations and prices, and the aggregate capital law of motion is an exact closed-form function of them, with no approximation of Φ . We derive the household policies and the saving rate, characterize the resulting *moment-recursive competitive equilibrium* (MRCE) and the properties the application uses, and show that the equilibrium is invariant to household forecasts, which is what makes it invertible on data.

3.1 Analytical household policies

The household's optimal policies solve in closed form.

Proposition 1 (Analytical household policies and wealth dynamics).

The optimal consumption and labor supply policies are:

$$c(x; X) = \frac{1 - \beta}{1 + \theta} (1 + r(X)(1 - \tau)e)a + \left(\frac{1 - \beta}{1 + \theta} a\xi + \kappa \right) w(X)(1 - \tau)Z_L z \quad (12)$$

$$n(x; X) = \max \left\{ \kappa(1 + \theta) + \xi a - \frac{\theta c(x; X)}{w(X)(1 - \tau)Z_L z}, 0 \right\} \quad (13)$$

The household's chosen saving is $\hat{a}'(x; X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi)$. The actual law of motion for wealth, after the unanticipated government transfer, is:

$$a'(x; X) = \hat{a}'(x; X) + T(X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi) + T(X). \quad (14)$$

Proof. The budget constraint can be rewritten as:

$$(1 + \theta)(c - \kappa Z_L z w(X)(1 - \tau)) + a' = ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi) a. \quad (15)$$

Under log utility, the standard result of [Brock and Mirman \(1972\)](#) yields $a' = \beta a R_i$. The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T(X)$. ■

Consumption is linear in wealth with source-dependent MPCs. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component (enabling aggregation) and an additive transfer (ensuring stationarity).

3.2 The saving rate

The saving rate has a closed-form characterization.

Proposition 2 (Analytical saving rate).

The saving rate $s(x; X) = \frac{a'(x; X) - a}{y(x; X)}$, where $y(x; X) = w(X)(1 - \tau)Z_L z n(x; X) + r(X)(1 -$

$\tau)ea + T(X)$ is after-tax income plus transfer, satisfies:

$$s(x; X) = \frac{a[\beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1] + T}{\tilde{w}Z_L z \kappa \theta + a \left[\frac{\tilde{w}Z_L z \xi(1 + \theta \beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e \right] + T}, \quad (16)$$

where $\tilde{r} = r(X)(1 - \tau)$ and $\tilde{w} = w(X)(1 - \tau)$, with the following properties:

$$(i) \lim_{a \rightarrow \infty} s(x; X) = \frac{\beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1}{\frac{\tilde{w}Z_L z \xi(1 + \theta \beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e}$$

(ii) $\lim_{a \rightarrow 0} s(x; X) > 0$ due to the transfer $T(X) > 0$

(iii) If $\beta R_i > 1$ (so the numerator coefficient $\alpha \equiv \beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1 > 0$) and the transfer is small relative to the subsistence term, $T < K_d/(\gamma/\alpha - 1)$ with $K_d \equiv \tilde{w}Z_L z \kappa \theta$ and γ the denominator coefficient in (i), then $\frac{\partial s}{\partial a} > 0$ and $\frac{\partial^2 s}{\partial a^2} < 0$. Both conditions hold throughout the calibrated region of Section 4 (where $\tilde{r} + \tilde{w}Z_L z \xi > (1 - \beta)/\beta$ in every period and $T \approx 0.001 K \ll K_d$); they may fail outside it.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive at zero wealth (due to T) and, under the dynamic-efficiency and small-transfer conditions of property (iii), increasing and concave in wealth over the calibrated region, matching the pattern documented by [Dynan et al. \(2004\)](#). Concavity generates mean reversion in relative wealth: as wealth rises, the subsistence floor absorbs a diminishing share and the saving rate converges to a constant.

Non-binding borrowing constraint An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint).

The household's chosen saving satisfies $\hat{a}'(x; X) = \beta a R_i(x; X) \geq 0$ for every (a, z, e, X) with $a \geq 0$, with equality only on the measure-zero set $\{a = 0\}$. The constraint $a' \geq 0$ is therefore slack in the household's optimization problem, independent of the transfer T .

Proof. By Proposition 1, $\hat{a}'(x; X) = \beta a R_i$ with $R_i = 1 + \tilde{r}(X)e + \tilde{w}(X)Z_L z \xi$. The labor-side term $\tilde{w}Z_L z \xi \geq 0$, and the capital-side term $1 + \tilde{r}e \geq 0$ whenever the after-tax return does not wipe out more than 100% of capital in one period, a mild condition satisfied throughout our calibration. With $\beta > 0$ and $a \geq 0$, the product is non-negative. The transfer T does not enter the household's Kuhn–Tucker condition because households are unaware of T when choosing a' ; it is added after the decision and only further raises the realized a' . ■

Slackness in the optimization problem is the structural property that keeps the policy globally linear and enables the analytical aggregation of Theorem 1. It rests on log preferences, which give a constant saving share β out of effective wealth aR_i , and on non-negativity of the effective return, not on the size of fiscal transfers.

3.3 Key equilibrium properties

The linear policies collapse the infinite-dimensional distribution to a finite moment vector. We state the properties the application relies on and relegate the formal definition, the existence proof, and the supporting derivations to Appendix G.

Sufficiency. The distribution Φ enters every contemporaneous equilibrium object only through $\mathcal{M} = [K, \text{cov}(z, a), \text{cov}(e, a)]$: aggregate capital and the two sorting covariances, between earnings ability and wealth and between portfolio returns and wealth. Given $\mathcal{M}$ and the aggregate state (A, Z_L, β) , all individual allocations and all prices (r, w, L) are determined without further reference to Φ . This defines the *moment-recursive competitive equilibrium* (MRCE), which exists uniquely (Appendix G.1).

Closed-form aggregate dynamics. Aggregate capital obeys an exact closed-form law of motion,

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi \text{cov}(z, a) + \tilde{r} \text{cov}(e, a)] + T, \quad (17)$$

where the after-tax prices $\tilde{r} = r(1-\tau)$, $\tilde{w} = w(1-\tau)$ and effective capital $K_{eff} = K + \text{cov}(e, a)$ set factor prices and aggregate labor:

$$\begin{aligned} r &= \alpha A (L/K_{eff})^{1-\alpha} - \delta, & w &= (1-\alpha)A (K_{eff}/L)^\alpha, \\ L &= Z_L\kappa + \frac{Z_L\xi(1+\theta\beta)(K + \text{cov}(z, a))}{1+\theta} - \frac{\theta(1-\beta)}{(1+\theta)\tilde{w}} [K(1+\tilde{r}) + \text{cov}(e, a)\tilde{r}]. \end{aligned} \quad (18)$$

Equation (17) separates a representative-agent channel $(1 + \tilde{r} + \tilde{w}Z_L\xi)K$ from a *return-sorting* channel $\tilde{r} \text{cov}(e, a)$ and a *productivity-sorting* channel $\tilde{w}Z_L\xi \text{cov}(z, a)$; the same shock moves the economy differently depending on how strongly wealth is sorted with returns and with productivity. The aggregate consumption and employment functions admit the same representative-agent-plus-sorting decomposition, and the equilibrium interest rate falls in return-wealth sorting, $\partial r / \partial \text{cov}(e, a) < 0$ (Appendix G.6 and G.7).

Ergodicity and nested benchmarks. The wealth recursion $a' = \beta a R_i + T$, with $R_i = (1 + \tilde{r}e) + \tilde{w}Z_L z \xi$, is a Kesten process: a positive additive intercept T on a multiplicative root delivers a unique stationary distribution with Pareto tails, and the dominant eigenvalue governing wealth-distribution dynamics is near unity (empirically ≈ 1.01 at the calibration), so mean reversion is slow and the sorting channels propagate shocks persistently (Appendix G.4). Muting return heterogeneity ($e = 1$) reduces $\mathcal{M}$ to $[K, \text{cov}(z, a)]$ and recovers the analytical Krusell and Smith (1998) economy; muting all heterogeneity ($e = z = 1$) reduces it to K and returns the Brock and Mirman (1972) growth model (Appendix G.5).

Closure. While $\mathcal{M}$ is sufficient contemporaneously and K' is exact, the forward evolution of the covariances depends on higher-order cross-moments outside $\mathcal{M}$. The quantitative analysis therefore tracks the full cross-section by simulation, using the exact household policies and the exact K' recursion and computing the covariances from the realized distribution (Appendix G.3).

3.4 Forecast invariance

A distinctive feature of the MRCE is invariance to household forecasts about aggregate dynamics, which removes the fixed-point problem that dominates standard heterogeneous-agent computation. In the forecast-invariant recursive competitive equilibrium (FRCE), households solve their problem under *arbitrary* beliefs about the aggregate laws of motion (Appendix G.2).

Theorem 1 (Forecast invariance).

In the FRCE, the law of motion Γ_Φ is unaffected by households' expectations about future aggregate states.

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L z \xi) + T(X)$ depends only on current prices, not on forecasts: under log utility the income and substitution effects of future return changes exactly offset. The FRCE therefore coincides with the MRCE (Appendix G.2), and the equilibrium is computed by forward simulation from any initial distribution Φ_0 and aggregate sequence $\{A_t, Z_{L,t}, \beta_t\}$, with no fixed-point iteration, no assumption on expectations, and no imposed stationarity. This is also what enables the inversion: structural shocks are recovered from data without taking a stand on expectation formation.

Corollary 2 (No terminal condition).

The equilibrium path generated by the forward simulation is fully determined by Φ_0 and

the aggregate sequence $\{A_t, Z_{L,t}, \beta_t\}$, and depends on no terminal date, terminal value, or transversality condition.

Proof. By Theorem 1, the saving policy $a' = \beta a R_i + T$ depends only on the current state; given Φ_t and X_t , prices clear and Φ_{t+1} follows household-by-household, so iterating from Φ_0 pins down the path as a first-order forward recursion with no terminal condition. ■

This marks a sharp departure from forward-looking equilibrium concepts. A standard recursive competitive equilibrium solves a two-point boundary-value problem: its Euler equation pins down behavior only with a terminal (or stationary) endpoint that selects the saddle path, so conventional methods need either a terminal steady state or an infinite forecast of future prices. Under log utility those forces cancel, the saving share is the current β_t regardless of what follows, and the equilibrium becomes an *initial-value* problem integrated forward like a physical law of motion. Practically, the data need not run to a terminal steady state: the sample can end mid-transition in 2019 with no assumption about the post-sample future.

Trend, cycle, and the raw-data inversion. The same property lets the framework invert the original aggregates as they are. Conventional structural models are defined relative to a stationary equilibrium or a deterministic trend, so confronting them with data first requires detrending or filtering the raw series into a cycle-only object. The MRCE instead absorbs trend growth and business-cycle fluctuations jointly into the recovered sequence $(A_t, \beta_t, Z_{L,t})$ within one equilibrium system, with no separate treatment of the long run and the cycle and no assumption that the economy ever reaches a stationary distribution. This is why the inversion runs directly on the raw U.S. 1950–2019 series and reproduces them period by period.

Remark 1 (Saving share under time-varying β).

Since β is an element of the aggregate state, it may vary over time. The value function is then $v(a, z, e; X) = \zeta \log a + B(z, e, X)$ with $\zeta = (1 + \theta) + \beta \mathbb{E}[\zeta']$; under forecast invariance households treat the current β as permanent, so $\zeta = (1 + \theta)/(1 - \beta)$ and the saving share is $s = \beta$. All results hold with the current β drawn from X , with no additional state variables. See Online Appendix A.8 for the ζ recursion.

Reconciling with the expectations literature. The Bellman decomposition $v(x; X) = u(x; X) + \beta \mathbb{E}[v(x'; X')]$ isolates a forward-looking term whose movements can be read either as changes in patience β or as changes in the expected continuation value $\mathbb{E}[v]$; with β free, the two are not separately identified from current behavior. The expectations literature breaks

the tie by fixing β and inferring $\mathbb{E}[v]$ (Moll, 2025; Bhandari et al., 2025); the MRCE breaks it the other way, making $\mathbb{E}[v]$ irrelevant to policies (Theorem 1) and loading the forward-looking content onto a directly identified β , recovered as β_t from aggregates (Section 4.2) and as β_i from PSID and SCF (Section 4.3). This is what makes β *semi-structural* in our setting: forecast invariance removes the expectational fixed point, but the same cancellation bundles patience and expectation into the recovered β , which is therefore structural only up to an expectational wedge. The bundling is a feature, not a defect of measurement: it is what lets a single, interpretable scalar carry the belief content that Moll (2025) argues is otherwise intractable, and it leaves a sharper structural decomposition of β , isolated from the sorting mechanism that drives our results, available to future work (Section 4.4). The strong negative correlation $\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.858$ in Section 4 then reads as the macro signature of a precautionary expectations channel, with the cross-sectional dispersion of β_i (Section 5.2.1) its analogue.

3.5 What the analytical structure buys for the inversion

The MRCE is not the only model that can be confronted with data. A forward-looking heterogeneous-agent model can also be inverted, by specifying a belief rule and a terminal condition, solving it numerically, and searching over parameters in an outer loop until simulated and observed moments coincide; filtering and likelihood-based estimation do exactly this. The contribution here is not that inversion is otherwise impossible but that the analytical structure makes it direct. Contemporaneous allocations are closed-form in $(\mathcal{M}, A_t, Z_{L,t}, \beta_t)$ and computed from the realized cross-section rather than a truncated aggregate state (a moment, projection, or linearization approximation (Krusell and Smith, 1998; Reiter, 2009; Winberry, 2018; Ahn et al., 2018; Auclert et al., 2021)), so the structural drivers are read off the data by inverting explicit equations, with no outer estimation loop, no re-solving, and no truncation of Φ . Two features of the model remove modeling choices that an inversion otherwise has to make. Forecast invariance means the recovered shocks do not depend on an assumed belief rule, so the inversion takes no stand on expectation formation; and the absence of a terminal condition (Corollary 2) means the sample need not run to a steady state. The result is a fast, data-anchored inversion that fits output, capital, and hours period by period and runs at the household level wave by wave, where an outer-loop estimator on a forward-looking model would be far more costly and would inherit its belief and terminal-condition assumptions.

The exact period-by-period fit is what makes the counterfactuals in Section 5.1.4 clean comparisons rather than a source of causal identification on its own. Because the factual

baseline reproduces the data exactly, the only difference between the factual and counterfactual paths is the policy change being studied, so the comparison is not confounded by baseline misfit, as it would be for a model matched to long-run averages. The causal content of the counterfactual still rests on the maintained assumptions of the structural model, in particular that the recovered (z_i, e_i, β_i) are policy-invariant (Section 5.2.2 tests this directly); the exact fit sharpens the comparison, it does not by itself establish causality.

4 Structural inversion at two scales

Section 3 showed that the equilibrium is closed-form and forecast-invariant. This section turns that property into a measurement device. Because every contemporaneous allocation and price is an explicit function of the structural drivers and the distributional state, the equilibrium equations can be read backwards: given the data, they return the structural objects whose general-equilibrium path reproduces that data exactly, with no forward solution, no estimation loop, and no assumption about expectations.

The inversion runs at two layers. At the aggregate layer, time series on output, capital, and hours map period by period into a few aggregate drivers; at the household layer, taking the aggregate-inverted prices as given, each surveyed household’s cross-sectional observables map into its own parameters, and the two nest, with the household parameters aggregating back to the macro objects. Which objects to read off is itself a choice: the equations admit several admissible inversions, and we adopt one natural baseline, recovering productivity, patience, and labor efficiency $(A_t, \beta_t, Z_{L,t})$ at the aggregate layer and the discount factor, earnings ability, and return type (z_i, e_i, β_i) at the household layer, holding the rest at calibrated values. Section 4.4 lays out the full menu of admissible choices and why this baseline is natural; here we implement it.

We begin with the data and calibration, then implement the aggregate and household inversions, and validate both.

4.1 Data and calibration

Data. The aggregate inversion uses annual U.S. data from 1950 to 2019 on real per-capita GDP, the per-capita capital stock, and aggregate hours, with the time-varying capital share α_t and depreciation rate δ_t drawn from the Penn World Tables and BEA fixed-assets accounts. The household inversion uses two surveys: the PSID (biennial, 2005–2019), whose panel dimension identifies the saving margin, and the Survey of Consumer Finances (SCF, 1992–2019), whose large repeated cross-sections sharpen the static margins. The three aggregate

series enter the aggregate equilibrium equations; each surveyed household's wage, capital income, wealth, and hours enter the closed-form policies as measurement equations. The survey-specific treatment is detailed in Section 4.3.

Calibration. The calibration is otherwise deliberately conventional. The analytical expressions carry over with α replaced by α_t period by period, since each period's equilibrium is contemporaneous. The reference annual discount factor $\beta = 0.96$ is the standard value (Cooley and Prescott, 1995). The three preference parameters (κ, θ, ξ) are calibrated jointly to three U.S. moments, in the spirit of Cooley and Prescott (1995) and Chang and Kim (2007). The leisure weight $\theta = 4$ delivers an intensive-margin Frisch elasticity of approximately 0.5 at the calibrated average-hours share, the value most common in the macro labor literature (MaCurdy, 1981; Keane and Rogerson, 2012; Chang and Kim, 2007; Heathcote et al., 2017); Online Appendix E.12 reports robustness. The base time endowment κ targets an average market-hours share near one third, the convention in Aiyagari–Bewley economies with endogenous labor (Chang and Kim, 2007; Heathcote et al., 2017). The wealth-leisure complementarity ξ is disciplined by the cross-sectional gradient of hours in wealth measured in the quasi-experimental wealth-effect literature (Imbens et al., 2001; Cesarini et al., 2017), which in our annual calibration implies a steady-state saving-to-labor-income ratio of 0.33, consistent with U.S. NIPA averages. Idiosyncratic earnings risk follows Storesletten et al. (2004), and idiosyncratic return risk follows Fagereng et al. (2020). The transfer share η is set so steady-state wealth concentration matches a U.S. wealth Gini near 0.80 (Díaz-Giménez et al., 2011; Kuhn et al., 2020b).

4.2 Aggregate inversion

The aggregate inversion uses the three equilibrium equations, the production function, labor supply, and capital accumulation:

$$Y_t = A_t K_{eff,t}^\alpha L_t^{1-\alpha}, \quad (19)$$

$$L_t = Z_{L,t} \kappa + \frac{Z_{L,t} \tilde{\xi}_t (1 + \theta \beta_t)}{1 + \theta} (K_t + \text{cov}_t(z, a)) - \frac{\theta(1 - \beta_t)}{(1 + \theta) \tilde{w}_t} (K_t(1 + \tilde{r}_t) + \tilde{r}_t \text{cov}_t(e, a)), \quad (20)$$

$$K_{t+1} = \beta_t \left[(1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + T_t, \quad (21)$$

where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (Online Appendix B). Note that equation (20) is implicit in L because prices depend on L through

Table 1: Calibrated parameters

Parameters	Description	Value	Source
Data-anchored (time-varying)			
α_t	Capital share	0.351–0.414	Penn World Tables
δ_t	Depreciation rate	0.037–0.058	BEA Fixed Assets
A_t	Aggregate TFP	Inverted	GDP, capital, hours
β_t	Discount factor	Inverted	Capital accumulation
$Z_{L,t}$	Labor productivity	Inverted	Labor supply
Calibrated (fixed)			
β	Reference discount factor	0.9600	Annual standard
θ	Labor disutility	4.0000	Frisch elasticity ≈ 0.5
κ	Common time endowment	0.2700	Avg. market hours share
ξ	Wealth–leisure complementarity	0.0213	Wealth-effect gradient on hours
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Idiosyncratic risk			
ρ_z	Prod. persistence	0.9500	Storesletten et al. (2004)
σ_z	Prod. volatility	0.1200	Storesletten et al. (2004)
ρ_e	Return persistence	0.9990	Fagereng et al. (2020)
σ_e	Return volatility	0.0100	Fagereng et al. (2020)

Notes: Calibration is annual. The table reports the baseline used in the U.S. inversion and in the distributional simulation.

the marginal products (Online Appendix A proves unique existence). The distributional covariances entering $K_{eff,t}$ are obtained from a preliminary two-series inversion that provides the moment-recursion covariance path (Online Appendix E). Given the distributional state, these equations recover $(A_t, \beta_t, Z_{L,t})$ period by period. TFP is identified from production, the saving-margin wedge β_t from capital accumulation, and the labor-efficiency wedge $Z_{L,t}$ from labor supply.

The labor block is identified recursively. The observed hours series enters the production function (19) as the efficiency-weighted labor input, $L_t = Z_{L,t} \int z n d\Phi$, measured in model units; given $K_{eff,t}$ and the capital share, A_t is the production residual that makes (19) hold, and the factor prices (r_t, w_t) follow from the marginal products.

With A_t , prices, β_t , and the distributional state in hand, the labor-supply equation (20) is one equation in the single remaining unknown $Z_{L,t}$, which it pins exactly: $Z_{L,t}$ is the labor-supply wedge that reconciles the observed labor input with the households’ optimal hours at the recovered prices. It is therefore identified from labor supply given A_t and prices,

not separately from the production residual. This makes $Z_{L,t}$ a wedge in the same statistical sense as the labor wedge of [Chari et al. \(2007\)](#); what distinguishes it is structural, not a matter of separate identification, as we discuss next.

The two equations are solved in sequence (production for A_t , then labor supply for $Z_{L,t}$), with the mutual dependence of β_t and $Z_{L,t}$ through the wage resolved by a short inner fixed point. The distribution is then simulated forward using the exact household policy.

The recovered shocks have distinct structural interpretations. A_t captures neutral total factor productivity. β_t captures time-varying patience, or equivalently demand-side forces operating through the saving margin. $Z_{L,t}$ is recovered as a labor-supply wedge, but it is a wedge with a structural place in the model rather than a free residual appended to it. Although it is backed out from the labor equation just as the labor wedge of [Chari et al. \(2007\)](#) is, it differs in three respects. First, it has a direct structural interpretation in the primitives of Section 2: $Z_{L,t}$ is the aggregate labor-efficiency index, and its movements map to shifts in labor-force composition, matching quality, or utilization intensity. Second, it is a parameter of the household’s effective wage, not of the production function, so it scales individual decisions rather than rescaling aggregate output as TFP does; this is what makes its time path distinguishable from A_t rather than absorbed into it. Third, it has distributional consequences: $Z_{L,t}$ enters the saving policy and equilibrium prices, so shifts in it alter the dynamics of wealth inequality, which a production-side labor wedge would not.

4.3 Household inversion

The same MRCE policies can be inverted at the household level. This subsection runs the inversion on PSID and the Survey of Consumer Finances (SCF), treating each survey as a sub-population that takes the macro-inverted GE prices as exogenous. The exercise identifies structural parameters (z_i, e_i, β_i) at the household level, provides an external validation of the macro β_t as a capital-weighted aggregate of micro β_i , and sharpens the distributional content of the framework with direct cross-sectional evidence on sorting.

PSID and SCF as sub-populations. A key methodological point is that aggregate prices (r_t, w_t) , the labor productivity index $Z_{L,t}$, the transfer T_t , and the growth-adjusted complementarity $\tilde{\xi}_t$ are taken from the macro inversion, not re-estimated from the household sample. PSID and SCF are not the universe: PSID’s wealth coverage excludes the very rich; SCF over-samples the wealthy through the list-sample but is cross-sectional. Re-estimating prices from a non-representative sample biases identification. Under the universe-level GE prices, each surveyed household solves the MRCE problem of Section 2, and the closed-form policies of Proposition 1 deliver direct measurement equations.

Measurement equations. For each household i in wave t , three observables identify three structural parameters in closed form, with no consumption data required and no numerical iteration.

Productivity z_i from the FTE wage. The model implies $W_i^d = \tilde{w}_t Z_{L,t} z_i$ where W_i^d is the after-tax full-time-equivalent wage. Pre-tax both sides cancel the labor tax, giving $z_i = W_{\text{nom},i}/(w_t Z_{L,t})$. We renormalize z_i within wave so that $E_w[z_i \mid \text{worker}] = 1$, absorbing the universe-vs-sample wage-level gap that arises because the macro w_t averages over the total population while the survey conditions on workers.

Return factor e_i from capital income. The model implies $\text{KI}_i = r_t e_i a_i$ in pre-tax terms (or $\tilde{r}_t e_i a_i$ post-tax), so $e_i \propto \text{KI}_i/(r_t a_i)$. We compute $e_i^{\text{raw}} = \text{KI}_i/(r_t a_i)$ and renormalize to $E_w[e_i \mid \text{KI} > 0] = 1$ within wave; the renormalization absorbs both the gross-vs-net-tax convention $(1 - \tau_K)$ and any sample-level reporting biases. Households reporting zero capital income (typically those with wealth concentrated in housing or retirement accounts that do not generate reported interest or dividends) have e_i imputed to 1; the imputation rate is 50–65% in PSID and 30–55% in SCF.

Discount factor β_i . Our baseline recovers β_i from the static intratemporal labor FOC in *both* surveys, so the two are identified the same way. Combining the labor FOC with the consumption equation gives

$$\beta_i^{\text{stat}} = 1 - \frac{(1 + \theta) \tilde{w}_t Z_{L,t} z_i (\kappa + \tilde{\xi}_t a_i - n_i)}{\theta R_i a_i}, \quad R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i, \quad (22)$$

where n_i is the household's labor share. It uses only contemporaneous hours, wage, and wealth, so it applies to any cross-section and requires no panel dimension.

PSID's panel dimension permits a second, dynamic identification. Iterating the saving policy $a_{t+1} = \beta a_t R_t + T_t$ over the biennial wave gap and assuming within-window constant prices yields $a_{t+2} = \beta_i^2 a_{i,t} R_i^2 + T_t(1 + \beta_i R_i)$, a quadratic in β_i with closed-form positive root

$$\beta_i^{\text{dyn}} = \frac{-T_t + \sqrt{T_t^2 + 4a_{i,t}(a_{i,t+2} - T_t)}}{2a_{i,t}R_i}. \quad (23)$$

Because (23) reads β_i off the two-year wealth change, it inherits the large transitory volatility of household wealth and yields a far noisier estimate than the static route (weighted $P_{90} - P_{10}$ of 1.57 versus 0.60 on the same households; Appendix I.3). We therefore take the static FOC as the baseline for both surveys and use the dynamic route in the appendix, where it also

frees the labor FOC to recover a household leisure weight θ_i (Appendix G.8). Both formulas are exact closed forms in the inverted parameters; the θ does not enter (23), and consumption does not enter either.

4.4 What the inversion can and cannot recover

The household inversion above recovers (z_i, e_i, β_i) and holds (κ, θ, ξ) at their calibrated values. This split is neither arbitrary nor the only admissible one. This subsection sets out which alternative choices the closed-form policies permit, which they forbid, and in what sense the baseline choice is innocuous. Two layers must be kept apart: the *measurement* layer, where each household satisfies its own first-order conditions given GE prices, and the *aggregation* layer, where the three-moment MRCE collapse constrains which parameters may differ across households.

The measurement system is block-triangular. Given the macro-inverted prices $(\tilde{r}_t, \tilde{w}_t, Z_{L,t}, T_t)$, each observable maps to one closed-form equation, and each structural object appears in only a subset of them (Table 2). The full-time-equivalent wage contains labor productivity alone; capital income contains the return factor alone; the two-period wealth change contains (β_i, ξ) once (z_i, e_i) are known; and the intratemporal labor condition contains $(\theta, \kappa, \xi, \beta_i)$. Because the first two equations each hold a single object with no preference parameter, the system is triangular: z_i and e_i are pinned first and free of preference contamination, and only then do the preference parameters enter, through one dynamic equation (saving) and one static equation (labor).

Table 2: Identification map for the household inversion

Closed-form equation	Param. present	Baseline use
Wage: $W_i^d = \tilde{w}_t Z_{L,t} z_i$	z_i	$\rightarrow z_i$
Capital income: $KI_i = \tilde{r}_t e_i a_i$	e_i	$\rightarrow e_i$
Labor FOC: $n_i = \kappa(1 + \theta) + \xi a_i - \frac{\theta c_i}{\tilde{w}_t Z_{L,t} z_i}$	$\theta, \kappa, \xi, \beta_i$	$\rightarrow \beta_i$ (both surveys)
Saving: $a_{i,t+2} = \beta_i^2 a_{i,t} R_i^2 + T_t(1 + \beta_i R_i)$	β_i, ξ	$\rightarrow \beta_i$ (PSID alternative); frees θ_i

Notes: Prices $(\tilde{r}_t, \tilde{w}_t, Z_{L,t}, T_t)$ are taken from the macro inversion. z_i and e_i are recovered first from the top two equations, which contain no preference parameter; the remaining unknowns $(\beta_i, \theta, \kappa, \xi)$ enter only the static (labor) and dynamic (saving) equations. The baseline recovers β_i from the labor FOC in both surveys. PSID's panel additionally supplies the saving equation, an alternative β_i route (Appendix I.3) that, when used for β_i , frees the labor FOC to recover a household leisure weight θ_i (Appendix G.8); the cross-sectional SCF provides no additional identifying equation.

Locked: z_i and e_i . No preference parameter appears in the wage or capital-income equation, so neither z_i nor e_i can be traded for θ , κ , or ξ . One cannot, for instance, invert θ “instead of” z from earnings, because θ does not appear there. Recovering (z_i, e_i) is therefore directly identified: declining to do so would discard the most direct, preference-free measurement the data provide.

Swappable: two observational equivalences. Two pairs are equivalent within a single equation, so which member is inverted and which is fixed is an identifying normalization, not a testable restriction. First, β_i and ξ both enter the saving equation through $\beta_i R_i$ with $R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i$; given (z_i, e_i) , the same wealth path reads either as discount-factor heterogeneity (invert β_i , fix ξ , our baseline) or as time-complementarity heterogeneity (invert ξ_i , fix β). Second, the return factor e_i and the complementarity ξ are both sources of dispersion in the saving multiplier R_i , one on the capital side and one on the time-saving side; the capital-income equation is what attributes residual return dispersion to e_i rather than ξ , and without it the two would be confounded.

Hours-dependent, hence PSID-only: θ_i and κ_i . The leisure weight θ and the base endowment κ appear only in the static labor equation (and, with consumption data, the consumption policy); they are absent from the wage, capital-income, and saving equations. Recovering either at the household level therefore needs hours data and an independent pin on β_i that frees the labor equation. PSID supplies both, since β_i comes from the wealth change and the hours equation is otherwise unused, which makes it possible to recover a household-level leisure weight θ_i as well; we develop that extension in Appendix G.8. The cross-sectional SCF cannot, because there the hours equation is already used to identify β_i . Within the labor equation, θ and κ enter jointly as $\kappa(1 + \theta)$, so hours identify one given the other; the exercise fixes κ and ξ .

Innocuousness, by layer. At the measurement layer the baseline split is innocuous for fit but not for interpretation: the equivalences above leave the recovered fit unchanged while re-labeling the economic source of heterogeneity, so reporting β_i rather than ξ_i , and fixing (θ, κ) , should be read as a normalization. At the aggregation layer the parameters are not interchangeable, and the reason is structural rather than conventional. The discount factor enters the saving and labor policies linearly and multiplying $a_i R_i$, so cross-household dispersion in β_i is summarized *exactly* by the single capital-weighted aggregate $\bar{\beta} \equiv \mathbb{E}_\Phi[\beta_i a_i R_i] / \mathbb{E}_\Phi[a_i R_i]$ already carried in the state vector (Section 2.2), with no new moments; this is why the framework admits a non-degenerate β_i distribution in the first place. The leisure weight is absent

from the saving policy, so heterogeneity in θ_i leaves the capital law of motion (17) untouched and would enter only the labor aggregate, costing at most one additional weighted moment. The endowment κ and the complementarity ξ instead multiply the idiosyncratic states (ξ multiplies z inside R_i), so making them household-specific introduces genuinely new cross-moments and enlarges $\mathcal{M}$ beyond three. This ordering, in which heterogeneity in β requires no additional moment, θ one additional moment, and ξ or κ genuinely new cross-moments, is what privileges (z, e) as the carried heterogeneities and $\bar{\beta}$ as the one preference aggregate the framework already tracks. A household-level θ_i is thus admissible as a measurement object, validated, like β_i , by checking that its appropriately weighted aggregate reproduces the calibrated value used in the macro labor equation.

Patience versus belief: the one composite the inversion cannot split. A last indeterminacy is of a different kind, because no normalization resolves it. Forecast invariance fixes the saving share at the current β regardless of what households expect, which is exactly what removes the expectational fixed point and makes the inversion well posed. The same cancellation, however, means current behavior cannot tell patience from expectation: a household that saves more because it discounts the future less and one that saves more because it is pessimistic about future returns are observationally identical. The recovered β_i is therefore a composite, structural patience times an expectational wedge, and every inversion in this paper reports the composite. Because the baseline reads β_i from the static labor FOC (22) in both surveys, the wedge enters the same way in each: the FOC substitutes in the *model's* consumption policy, so any gap between a household's true consumption and that policy, whether from a belief that differs from the model's or from any other unmodeled force, is absorbed into $\hat{\beta}_i$ rather than left as a residual. Separating patience from the wedge therefore requires a second, independent equation, and observed consumption is the natural one: the gap between measured c_i and the model's consumption policy at the recovered (z_i, e_i, β_i) identifies the wedge directly, as an added cross-sectional restriction in SCF and against the saving residual in PSID, whose panel already supplies the dynamic route (23) as a check. We leave this structural-plus-econometric decomposition of β to future work; the present analyses assume no separate belief channel and read β_i as the composite throughout.

4.5 Recovery and validation

The inversion fits the data exactly by construction, so it cannot be judged by in-sample fit. We discipline it three ways instead: a falsification test on simulated data, external indicators never used in identification, and cross-survey replication of the household inversion.

Falsification. A first check asks whether the inversion manufactures shocks that are not there. We simulate three Krusell–Smith economies driven by aggregate TFP alone (fixed labor; endogenous labor; and Frisch labor with adjustment costs) and apply the same inversion to the simulated (Y_t, K_t, L_t) . It recovers the true TFP process and leaves β_t and $Z_{L,t}$ essentially flat (Figure 1; the accuracy table is in Appendix H), so it does not mechanically create preference or labor shocks.

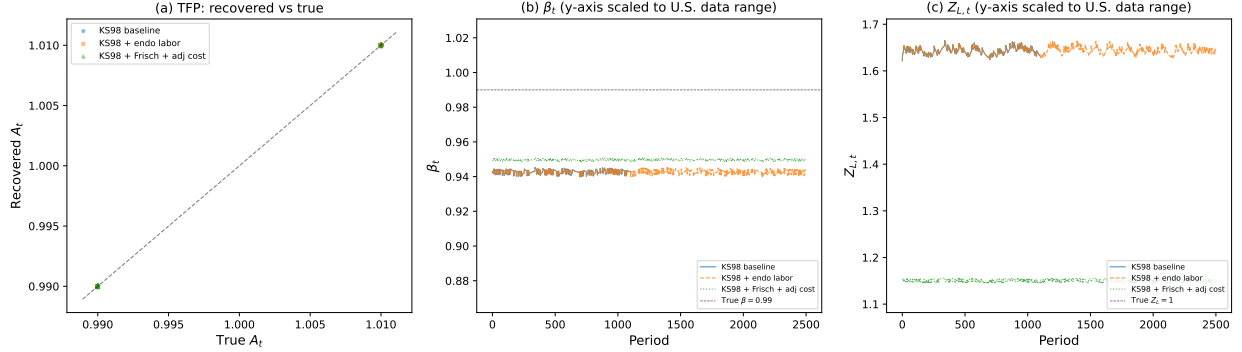


Figure 1: Krusell–Smith validation: recovered shocks from simulated data

Notes: Sequential inversion applied to simulated (Y_t, K_t, L_t) from three Krusell–Smith variants. The recovered TFP aligns with the true aggregate TFP process, while the recovered β_t and $Z_{L,t}$ series are nearly flat on the U.S. data scale.

Applied to U.S. data, however, the inversion produces substantial variation in β_t and especially $Z_{L,t}$: postwar capital accumulation is well approximated by a single TFP shock with near-constant preferences, but the labor market requires a second structural margin that a one-shock economy cannot generate.

Recovered series Figure 2 reports the recovered structural series. TFP grows from 0.807 in 1950 to 1.79 in 2019, tracing the postwar expansion, the 1970s slowdown, the IT revolution, and post-2008 stagnation. The saving-margin wedge β_t is persistent and countercyclical, with a secular decline from 1950 to 2019. The labor-efficiency wedge $Z_{L,t}$ fluctuates at business-cycle frequencies and captures movements in effective labor input beyond measured hours.

The MRCE-implied TFP tracks the representative-agent analogue closely at business cycle frequencies and correlates strongly with the Fernald utilization-adjusted series (Online Appendix E.1).

External discipline. Neither $Z_{L,t}$ nor the secular path of β_t is targeted, yet both line up with outside indicators. The labor-efficiency wedge co-moves with the employment-population ratio and labor-force participation, the extensive margins it is meant to capture,

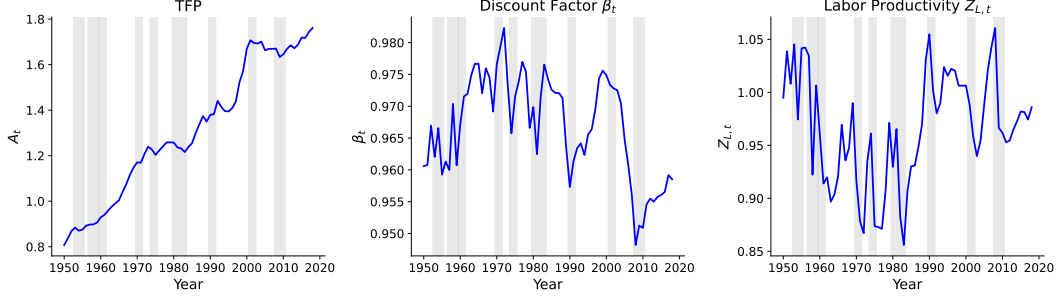


Figure 2: Recovered structural series from the data-anchored inversion

Notes: Top: aggregate TFP. Middle: saving-margin wedge β_t . Bottom: labor-efficiency wedge $Z_{L,t}$. The inversion uses observed output, capital, and hours. Shaded areas indicate NBER recession dates.

and runs *opposite* to the BLS labor-composition index, which rises in recessions as low-productivity workers are shed; the sign reversal confirms Z_L as a utilization-and-matching margin rather than a demographic-composition index (table in Appendix H). The saving-margin wedge in turn tracks the personal saving rate, labor-force participation, and revolving consumer credit relative to GDP (Figure 3), consistent with reading β_t as a compact saving-side wedge that absorbs demographic, financial-liberalization, and precautionary forces.

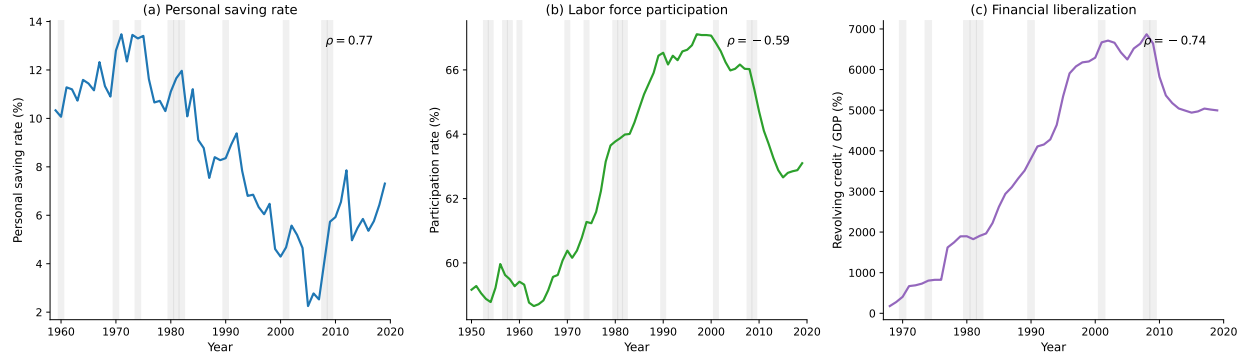


Figure 3: External discipline for the saving-margin wedge

Notes: External indicators from FRED, not used in the inversion. Panels show the U.S. personal saving rate, civilian labor-force participation, and revolving consumer credit as a share of GDP against the recovered β_t . Shaded areas indicate NBER recession dates.

Distinct shocks. The three recovered shocks are genuinely distinct rather than one residual relabeled. TFP is nearly orthogonal to both $Z_{L,t}$ and β_t , while labor efficiency and patience move together, $\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.858$, so labor-market deterioration coincides with higher saving (Figure 4). The cyclical content of this pairing, and a classification of postwar recessions by their driving shock, are developed in Section 5.

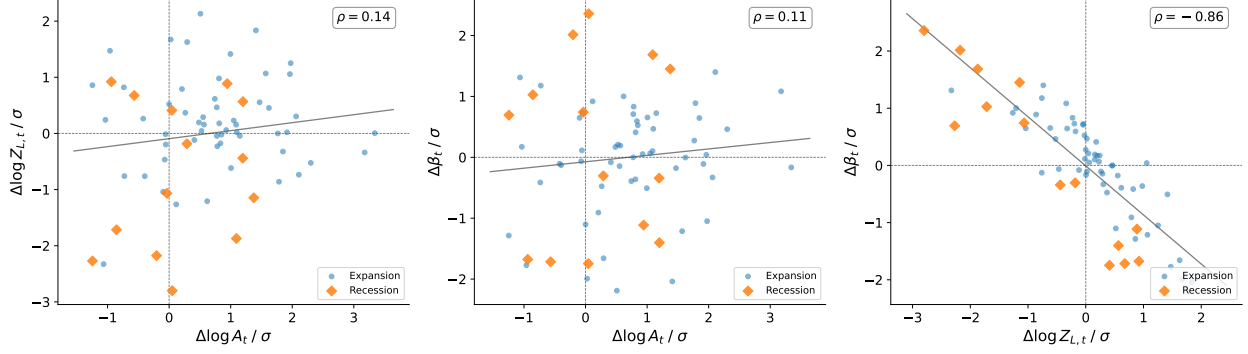


Figure 4: Pairwise scatter of recovered shock changes: the two-dimensional business cycle

Notes: Each point is one year. Blue circles are expansion years; orange diamonds are recession years. Changes are normalized by their unconditional standard deviations. The right panel shows the labor-saving tandem; the left and center panels show that TFP innovations are nearly orthogonal to the other two shocks.

Cross-survey replication of the household inversion. The household inversion is disciplined by replication across two independent surveys. We apply the static FOC (22) to both PSID (2005–2019, 8 biennial waves, $\sim 4,000$ households each) and SCF (1992–2019, 10 waves, 11,000–18,000 each; the 1989 wave is excluded as a methodology outlier), with β clipped to $[0.5, 1.5]$ to bound the formula’s tails; the dynamic route (23) is the PSID-only alternative (Appendix L.3). Table 3 reports population-weighted moments by wave.

Table 3: Recovered structural parameters: PSID and SCF, sample-mean across waves

Source	Years	N/wave	$\bar{\beta}_{\text{cap}}$	$\bar{\beta}_{\text{med}}$	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
PSID	2005–2019	4,109	0.936	1.012	−0.20	−0.31	+0.39
SCF	1992–2019	14,969	0.877	0.992	−0.28	−0.59	+0.22
Macro β (§4)	1950–2019	—	0.967	—	—	—	—

Notes: $\bar{\beta}_{\text{cap}} = E_w[\beta_i a_i R_i] / E_w[a_i R_i]$ averaged across waves. $\bar{\beta}_{\text{med}}$ is the wave-pooled median. Correlations are sample-mean across waves; standard deviations across waves are below 0.04 for $\rho(\beta, z)$ and below 0.08 for the gradients. Macro β is the 1950–2019 mean of the macro inversion; the PSID 2005–2019 sub-window mean is 0.956 and the SCF 1992–2019 sub-window mean is 0.962. SCF 1989 is excluded: it is a methodology outlier ($n=1,067$ vs. 11,000–18,000 in subsequent waves; $\rho(\beta_i, \log a_i)$ sign-flips to +0.25 versus −0.51 to −0.63 in all later waves) due to first-survey weighting that the AHEAD redesign fixed in 1992.

On disjoint samples spanning three decades, the two surveys agree closely on the typical household: the median β is 1.01 in PSID and 0.99 in SCF, within 2 pp, and the sign of every wealth-sorting gradient matches. The capital-weighted aggregates differ more, $\hat{\beta}_{\text{cap}} = 0.936$ in PSID and 0.877 in SCF, but for a known reason: the SCF list sample captures the ultra-wealthy, low- β top of the distribution that PSID misses, which pulls the SCF aggregate down. PSID’s aggregate therefore sits only 2 pp below the macro $\beta_t = 0.967$, while SCF

sits 8 pp below; the spread between them is survey wealth coverage, not identification, since both now use the same static FOC. This agreement on the typical household and on every gradient, across two independent pipelines that share only the model, is the household-scale counterpart of the aggregate falsification. We unpack the time-series structure in Section 5.2.1 and the macro-micro gap in Section 5.3.

Skew, dispersion, and the choice of summary. The recovered household parameters are all right-skewed, so we summarize them by medians throughout. Under the baseline static identification the two surveys are similarly dispersed in β_i (weighted $P_{90} - P_{10}$ of 0.60 in PSID and 0.36 in SCF); the return factor e_i is the most skewed, a capital-income-to-wealth ratio with a thin set of very large values. The dynamic (saving-equation) route is far noisier: on the same PSID households it inflates the β spread to 1.57, because it reads β_i off the two-year wealth change $a_{i,t+2}/a_{i,t}$, which is enormously volatile (its P_{10} to P_{90} runs from 0.28 to 4.84, wealth shrinking to a quarter or growing nearly fivefold in two years), and since $\beta_i \approx \sqrt{a_{i,t+2}/a_{i,t}} / R_i$ when the transfer is small, that volatility passes almost directly into β_i . A within-person split assigns 70% of the dynamic-route dispersion to transitory movement rather than persistent type (Table 7). The static FOC uses no two-period change and is correspondingly smooth, which is why we take it as the baseline and keep the dynamic route in the appendix.

The SCF clip at $[0.5, 1.5]$ has a complementary reading. Because the static FOC divides by wealth, $\beta_i = 1 - (1 + \theta) \tilde{w}_t Z_{L,t} z_i (\kappa + \tilde{\xi}_t a_i - n_i) / (\theta R_i a_i)$, it returns uninformative extremes at the two corners of the cross-section. Wealth-poor households drive the small denominator to zero, so β_i explodes upward: the 4.7% clipped to 1.5 have median wealth 0.13 against 12.5 in the interior. Very-high-wage, low-hours households make the labor-wedge term large, so β_i is pushed down: the 3.1% clipped to 0.5 have median z_i of 6.6 against 0.84. The clip caps these corners, where the identification is structurally weak, rather than letting them distort the cross-section; the interior 92% of the sample is untouched. Appendix 1.3 verifies both readings directly: it re-identifies β on the same PSID households by the static FOC, which collapses their dispersion to near the SCF level, and it decomposes the clipped SCF corners by wealth, hours, and occupation.

Because of this skew and noise, we summarize the household cross-section with medians throughout the household-scale results (Section 5.2.1), and we use a mean only for the capital-weighted $\bar{\beta}_{\text{cap}}$, the exact object that enters the macro state (Section 2.2). The median replicates closely across surveys even though the mean β_i do not: the wave-pooled medians agree within 2.3 pp ($\bar{\beta}_{\text{med}}$ of 1.015 in PSID and 0.992 in SCF, Table 3). The capital-weighted aggregates agree less closely ($\hat{\beta}_{\text{cap}}$ of 0.936 in PSID and 0.877 in SCF, a 5.9 pp difference),

because $\hat{\beta}_{\text{cap}}$ weights by $a_i R_i$ and so is dominated by the wealthy top that the two surveys cover differently (Section 5.3). The median life-cycle profiles of (z_i, e_i, β_i) likewise track each other across the two surveys (Section 5.2.1).

4.6 Data requirements and portability

The inversion is light on data, which makes it easy to apply to other datasets and other countries. This subsection states what a dataset must provide and points to data sources beyond the ones we use.

What the method needs. The aggregate inversion needs only annual series for output, the capital stock, and hours, together with the capital share, the depreciation rate, and tax rates. All of these come from standard national accounts. The household inversion needs three observables per household: earnings (for z_i), capital income together with wealth (for e_i), and either wealth observed two periods apart or hours worked (for β_i). The first route to β_i uses the saving equation; the second uses the labor first-order condition. No consumption data and no numerical solution are required at either scale.

Other micro data sources. We use PSID and SCF, but these are one choice among many. PSID does not cover the very rich, and SCF is a repeated cross-section rather than a panel. Administrative records would relax both limits. Tax and wealth registries, such as the Scandinavian population registers used in the return-heterogeneity literature (Fagereng et al., 2020), record earnings, capital income, and wealth for the entire population, with a panel dimension. Applying the inversion to such data would give a much larger and more representative sample, would cover the top of the wealth distribution that surveys miss, and would let the panel route to β_i be used for the whole population.

Cross-country application. The aggregate inversion is even easier to move, because output, capital, hours, the capital share, and the depreciation rate are available for many countries from sources such as the Penn World Tables. Feeding each country’s series through the same equations would recover a comparable panel of TFP, patience, and labor efficiency, and would allow a cross-country comparison of how growth, the cycle, and the distribution interact. We leave this extension to future work.

5 Macro and micro implications

The inversion produces two sets of structural objects: the aggregate shocks $(A_t, \beta_t, Z_{L,t})$ and the household parameters (z_i, e_i, β_i) . With these objects we can carry out several standard exercises without solving the model forward. At the aggregate level, we use the shocks to study the business cycle, to measure how growth and the wealth distribution interact, to forecast, and to run policy counterfactuals. At the household level, we use the parameters to describe the cross-sectional distribution of patience, ability, and returns, to see how it changes over the life cycle, and to test whether it is stable when policy changes.

This section presents these exercises. On the aggregate side, we classify postwar recessions by the shock that drove them, we describe a *sorting accelerator* in which TFP growth raises wealth concentration, we show that the model forecasts output and labor more accurately than a recursive AR benchmark, and we measure the distributional effects of four postwar capital-tax reforms. On the household side, we recover the joint distribution of (z_i, e_i, β_i) , we show where wealth sorting is strongest over the life cycle, and we test whether any household's β_i or z_i changes around the four reforms. This last test is a Lucas-critique test, and the recovered parameters pass it.

To keep the section readable, we present one main figure or table for each result and place the supporting material in the appendix. The main message is simple: a single inversion supports many uses, including business-cycle accounting, the sorting mechanism, forecasts, policy counterfactuals, and a household-level test, all within the same equilibrium framework.

5.1 What the aggregate inversion reveals

This half uses only the aggregate shocks $(A_t, \beta_t, Z_{L,t})$. We use them to study the business cycle, to generate distributional dynamics from a simulated cross-section, to forecast, and to run policy counterfactuals. None of these exercises uses the survey data. The survey data enter in the next half.

5.1.1 Aggregate dynamics and the structural recession classification

The analytical aggregate equations imply three quantitative lessons. First, representative-agent aggregation is an excellent approximation in the baseline calibration. Investment decomposes into a representative-agent channel, a return-sorting channel, and a productivity-sorting channel; the representative-agent channel accounts for 93.2 percent of mean investment. This explains why the Krusell–Smith approximation works so well even though the exact state contains additional covariances.

Second, the omitted covariances still create a state-dependent amplification channel, and the analytical structure makes the generalized impulse response especially transparent. In a conventional heterogeneous-agent model a generalized impulse response must be computed by Monte Carlo, averaging simulated paths with and without the shock over draws of the idiosyncratic and aggregate history, precisely because the response depends on the entire distribution at the date of the shock (Koop et al., 1996). Here the response is governed by the three sufficient moments $\mathcal{M}$: two economies with the same $\mathcal{M}$ have identical impulse responses regardless of the rest of the distribution, so the state dependence is summarized by a three-dimensional object rather than an infinite-dimensional one. We compute generalized impulse responses to a 2 percent negative TFP shock from the 1952 distribution, when wealth inequality is low, and from the 2019 distribution, when wealth inequality is high. The impact effect is 3 percent larger from the 2019 state (-2.29% versus -2.22%), and the cumulative 10-year output loss is modestly larger (-20.4% versus -20.2%). The magnitude is small at the conservative baseline calibration, but the sufficient statistic is transparent: amplification rises with return-wealth sorting, summarized entirely by $\text{cov}(e, a)$.

Third, the direct asset-pricing effect of return-wealth sorting is negligible at $\sigma_e = 0.01$. The interest-rate effect is approximately -0.001 basis points. This is a useful negative result. In the baseline calibration, return heterogeneity matters more as an amplification state variable than as a quantitatively important risk-premium channel. An Epstein–Zin extension, reported in the Online Appendix, is a more promising route for asset-pricing applications.

Structural recession classification. The exact inversion classifies recessions structurally, by reading off which shock moved, without narrative identification or external instruments. Table 4 reports recovered shocks around all seven NBER recessions in standard-deviation units. The classification rule assigns the label of the largest shock in $|\sigma|$ units; combined labels appear when multiple shocks exceed 1σ . The resulting typology is richer than supply-versus-demand: the 1957–58 and 1981–82 episodes combine labor and TFP channels, the 1973–75 episode features a large patience decline alongside modest TFP and labor movements, and the 2007–09 contraction is dominated by TFP. This heterogeneity in recession mechanisms illustrates the value of a three-shock framework.

5.1.2 The sorting accelerator and cross-sectional dynamics

The mechanism. The main distributional mechanism is a sorting accelerator. TFP growth raises returns. Because return differences are persistent, high-return households become richer over time, and a more concentrated wealth distribution then makes return-wealth

Table 4: Structural recession classification from the exact inversion

Episode	ΔY (σ)	$\Delta \log A$ (σ)	$\Delta \log Z_L$ (σ)	$\Delta \beta$ (σ)	Classification
1953–54 (Korean War)	−1.12	−0.86	−1.72	−0.09	Labor-driven
1957–58 (Eisenhower)	−1.16	+0.34	−2.98	+2.05	Labor-driven
1973–75 (Oil Crisis)	−1.27	−1.63	−2.33	−2.44	TFP/Labor
1980 (Volcker I)	−0.68	−0.04	−1.07	+0.74	Labor-driven
1981–82 (Volcker II)	−1.33	−1.14	−2.18	+0.34	TFP/Labor
1990–91 (S&L Crisis)	−0.69	+1.31	−1.26	−0.48	Labor-driven
2007–09 (Great Recession)	−2.07	−1.24	−2.27	−2.16	TFP/Labor

Notes: All entries are peak-to-trough changes in units of each variable’s unconditional standard deviation. Classification: “TFP-driven” when $|A|$ dominates; “Labor-driven” when $|Z_L|$ dominates; “Demand-driven” when $|\beta|$ dominates.

sorting matter more in later periods. The covariance $\text{cov}(e, a)$ is therefore very persistent: its dominant eigenvalue is above one in 100 percent of sample years, with mean 1.02. We isolate the mechanism with four counterfactuals (Figure 5): the baseline uses all recovered shocks, and the others switch off TFP growth or hold the aggregate shocks fixed. Both sorting channels grow along the baseline path. Without TFP growth, return-wealth sorting stops growing; without countercyclical dispersion, productivity-wealth sorting loses its recession spikes. Appendix I.1 gives the analytical decomposition of the two channels.

Who gains from growth. The same mechanism sends most of the gains from growth to the top. At the steady-state distribution, a one-percent rise in TFP raises the wealth of the top decile by 81.6 percent of the total gain, which is more than its 77.2 percent wealth share, because the two sorting channels add up. Along the historical path the top decile’s share of TFP-driven gains rises from about 67.3 percent in the 1950s to about 69.4 percent in the 2010s, as the wealth distribution becomes more concentrated. The full results by percentile and by decade are in Appendix I.1.

Matching the wealth distribution. When we simulate the model forward with the recovered shocks, it reproduces the level and the rise of U.S. wealth inequality through the rule $a' = \beta R_i a + T$ (Table 5). The wealth Gini rises from 0.827 to 0.876, driven by the high persistence of return-wealth sorting, while the transfer T keeps the bottom of the distribution in place. The model Gini rises more smoothly than the data. This gap is informative: it points to time-varying return dispersion and housing as the missing forces. The related cross-sectional results, namely the paths of consumption and income inequality, their behavior in recessions, the full 1980 and 2015 wealth distributions, and the closed-form aggregate consumption path, are in Appendix I.1.

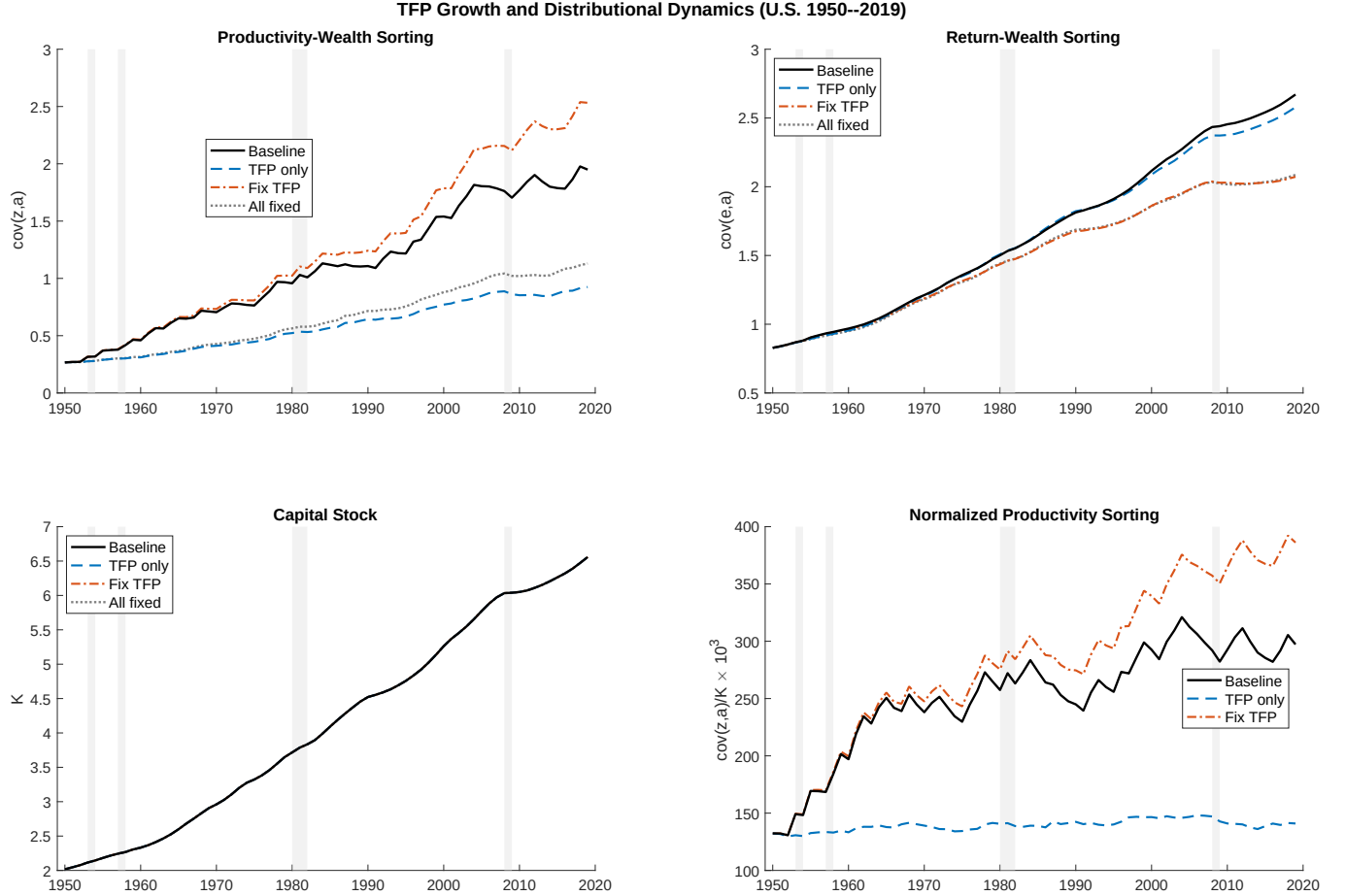


Figure 5: TFP growth and distributional dynamics

Notes: Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. Shaded areas indicate NBER recession dates.

Table 5: Wealth Gini: model vs. data

Year	Data	Model	Year	Data	Model
1953	0.73	0.830	1998	0.80	0.864
1962	0.78	0.838	2001	0.81	0.865
1968	0.77	0.843	2004	0.81	0.867
1983	0.77	0.855	2007	0.82	0.869
1989	0.79	0.858	2010	0.85	0.871
1992	0.78	0.860	2013	0.85	0.873
1995	0.78	0.862	2016	0.86	0.875

Notes: Data are SCF-based estimates reported by [Kuhn and Ríos-Rull \(2025\)](#). Model moments are from the historical simulation using exact household policies.

The effect of the sorting accelerator is largest when shocks are persistent. The next

subsection shows this with historical capital-tax reforms.

5.1.3 Conditional forecasting

A distinctive advantage of the MRCE is that conditional forecasting is computationally direct. In conventional heterogeneous-agent models, forecasting requires re-solving the household dynamic program or estimating a perceived aggregate law of motion, because policy functions depend on forecasts of future aggregate states. In the MRCE, household policies are backward-looking and linear in wealth. Given any future sequence for $(A_t, \beta_t, Z_{L,t})$, the model produces output, capital, labor, prices, consumption, investment, and the distributional state by forward simulation alone.

Rolling forecasts We exploit this property with a rolling forecast exercise. For each origin year $t \in \{1965, \dots, 2009\}$, the inversion recovers structural shocks from data available up to t , and the model is simulated forward holding shocks at their last recovered values. The resulting forecast jointly produces output, capital, labor, prices, consumption, investment, and the distributional state, all satisfying the production function, capital accumulation, and market clearing simultaneously.

Figure 6 compares the MRCE forecast to a recursive AR benchmark that forecasts the structural inputs using the same information set and aggregates them through the same production function. At both one- and three-year horizons, the MRCE has lower RMSE for output (3.1% vs. 7.0% at $h = 1$; 5.1% vs. 9.4% at $h = 3$) and labor (2.6% vs. 4.2%; 5.4% vs. 6.1%). For capital, the AR benchmark exploits the high persistence of the raw capital stock ($\rho = 0.99$) and is difficult to beat at short horizons. The MRCE's advantage is structural: it produces a jointly consistent forecast of all prices and quantities, while the AR inputs evolve independently and are not required to satisfy any equilibrium condition.

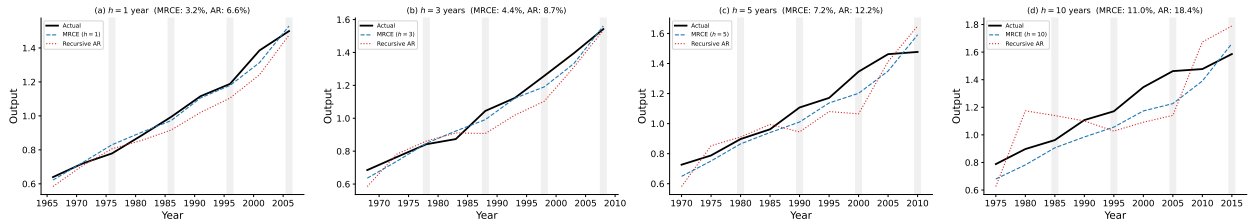


Figure 6: Rolling forecast errors by horizon

Notes: For each origin year, the model is simulated forward holding recovered shocks at their last values. The recursive AR benchmark forecasts structural inputs using the same information set and aggregates through the same production function. Panel titles report RMSEs. Shaded areas indicate NBER recession dates.

5.1.4 Distributional impact of tax reforms

The near-unit-root persistence of wealth sorting implies that even temporary changes in capital taxation can produce almost permanent shifts in the wealth distribution. We test this prediction using four historically abrupt capital tax changes from both directions of the political spectrum: two cuts (Reagan ERTA 1981, τ_K from 0.380 to 0.375; Bush EGTRRA 2001, τ_K from 0.395 to 0.355) and two increases (OBRA 1993 under Clinton, τ_K from 0.340 to 0.360; ATRA 2013 under Obama, τ_K from 0.278 to 0.310).²

For each episode, two full agent-based simulations (500,000 agents) use identical aggregate shocks and idiosyncratic histories. The counterfactual is *reform-only*: $\tau_{K,t}^{\text{cf}}$ equals the factual path minus the named reform’s reform-year shock from the reform date onward, leaving all other reforms intact. For example, the OBRA-only counterfactual subtracts +0.020 from $\tau_{K,t}$ for $t \geq 1993$ but keeps the Bush 2001 cut, ATRA 2013 increase, and TCJA 2017 cut in both factual and counterfactual paths. This isolates the marginal effect of each named reform rather than the cumulative effect of “no policy change after the reform date.” Figure 7 reports the cuts in the main text; the increases are shown in the same layout in Appendix I.2 (Figure I.14).

Several features of Figure 7 merit attention. First, the Gini gap opens immediately at each reform date and continues to widen for decades, confirming the near-unit-root prediction: once the wealth distribution is pushed toward greater concentration, persistence compounds the initial shift rather than allowing mean reversion. Second, the wage panel (row 4) reveals the growth side of the tradeoff: lower capital taxes stimulate capital accumulation, raising the marginal product of labor and hence equilibrium wages, so all households benefit from higher wages even as inequality rises. Third, the return-wealth covariance (row 5) diverges at each reform date, tracing the sorting mechanism directly: lower τ_K raises after-tax capital returns, which disproportionately benefit high-wealth households, accelerating $\text{cov}(e, a)$ growth.

Figure I.14 confirms that the same mechanism runs in reverse for tax increases. After OBRA 1993 raises τ_K by +0.020 (from 0.340 to 0.360), the factual wealth Gini falls relative to the reform-only counterfactual, the return-wealth covariance compresses, and output and wages contract. The Bush (−0.040) and ATRA (+0.032) reform-year shocks dominate the Reagan (−0.005) and OBRA (+0.020) shocks in magnitude, but cumulative distributional

²Annual effective tax rates ($\tau_{L,t}, \tau_{K,t}$) used in the inversion and counterfactuals are not directly observed at annual frequency. We construct them as a year-by-year stylized series consistent with the decade averages of McDaniel (2007) and the timing of major federal legislation (Korean War surtax, Kennedy 1964, ERTA 1981, TRA 1986, OBRA 1993, EGTRRA 2001, JGTRRA 2003, ARRA 2009, ATRA 2013, TCJA 2017). The full series is in `exact_ge_mapping/data/effective_tax_rates.csv`; see `build_tax_data.py` for the construction. Counterfactual conclusions about *differences* between factual and reform-only paths are robust to reasonable variation in the tax-rate series because both scenarios are subject to the same construction.

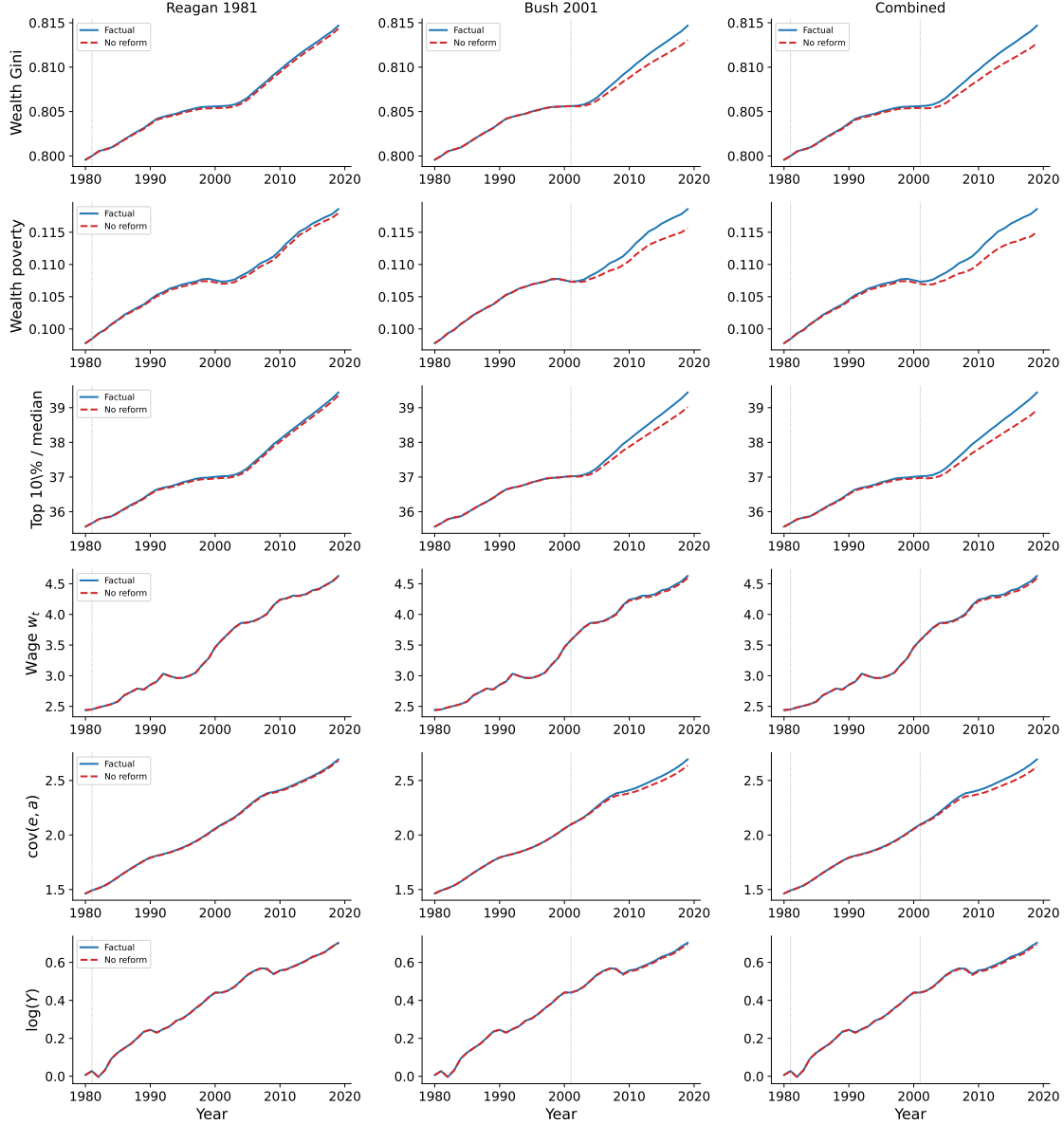


Figure 7: Macroeconomic impact of capital tax cuts (Reagan 1981, Bush 2001)

Notes: 500,000-agent simulation, 1980–2019. Columns: Reagan 1981, Bush 2001, and combined cuts. Rows show wealth Gini, wealth poverty (share of households with wealth below half the median), top 10%/median ratio, wage, return-wealth covariance, and output. Solid blue: factual; dashed red: counterfactual (no reform). Capital evolves endogenously; all GE prices re-solved each period.

effects are governed jointly by shock size and post-reform time horizon: OBRA 1993 has 26 post-reform years in the sample, ATRA 2013 only 6. The reform-only design isolates each reform from all subsequent policy: in particular, the OBRA panel is no longer contaminated by the post-1993 Bush 2001 and TCJA 2017 cuts.

Panel A of Table 6 quantifies the growth-versus-inequality tradeoff for the cuts under

the reform-only counterfactual. Stacking both reforms, the combined isolated effect by 2019 is a 1.01% output gain and 0.87% wage increase, offset by a 0.20 percentage point wealth Gini increase, a 1.30% rise in the top 10%/median ratio, and a 3.11% increase in the wealth-poverty rate. The income share of the top 1% rises by 0.40 percentage points while the bottom 50% loses 0.18 points. The magnitudes are smaller than under the freeze-pre-reform design (which conflates each reform with all subsequent policy), but the signs and rankings are the same: Bush EGTRRA delivers most of the cumulative effect because its reform-year cut (-0.040) is much larger than Reagan ERTA's (-0.005). Even under the cleaner interpretation, the bottom decile gains in absolute income levels but loses share, as the sorting mechanism channels the growth dividend toward the top.

Panel B of Table 6 reports the symmetric numbers for the two tax-increase episodes under the same reform-only design. Because OBRA 1993 lies 17 years before 2010 and ATRA 2013 lies 3 years *after* 2010, we use 2002 (9 years post-OBRA, before ATRA) and 2019 (terminal) as the horizons. Signs are uniformly negative for output, wage, and inequality measures, mirroring the cuts. By 2019 the combined isolated effect of both increases is -0.77% output, -0.67% wage, -0.15 pp wealth Gini, -1.00% top 10%/median, and -2.37% wealth poverty. The income share of the top 1% falls by 0.48 percentage points and the bottom 50% gains 0.21 points: the same sorting mechanism runs in reverse, redistributing toward the bottom when capital is taxed more heavily. OBRA 1993 carries most of the cumulative effect because it has 26 post-reform years in the sample versus only 6 for ATRA 2013.

The exercise establishes that the macro side's tax counterfactuals are not skewed toward one political direction: the same near-unit-root sorting mechanism produces decades-long distributional drift after both major cuts and major increases, with appropriately signed effects. Read together, the 1981, 1993, 2001, and 2013 episodes span the full postwar range of capital tax policy, and each leaves a structural footprint that the analytical framework traces directly. The magnitudes hold (z_i, e_i, β_i) fixed across the policy change; Section 5.3 discusses the scope of this maintained hypothesis.

5.2 What the household inversion reveals

This half uses the household parameters (z_i, e_i, β_i) recovered from PSID and SCF. In the previous half, the cross-section was simulated from calibrated processes. Here, the cross-section comes directly from the data. We describe its life-cycle pattern, how its sorting gradients change over time, and whether the parameters are stable when tax policy changes.

Table 6: Growth vs. inequality: capital tax reforms (reform-only counterfactuals)

<i>Panel A: Tax cuts</i>						
	Reagan 1981		Bush 2001		Combined	
Δ	2010	2019	2010	2019	2010	2019
<i>Growth (%)</i>						
Output	+0.13	+0.16	+0.44	+0.84	+0.58	+1.01
Wage	+0.12	+0.14	+0.42	+0.73	+0.54	+0.87
<i>Inequality measures</i>						
Wealth Gini (pp)	+0.03	+0.03	+0.09	+0.16	+0.12	+0.20
Top 10%/median (%)	+0.18	+0.22	+0.56	+1.07	+0.74	+1.30
Wealth poverty (%)	+0.46	+0.55	+1.47	+2.59	+1.90	+3.11
<i>Income share shift (pp)</i>						
Top 1%	+0.04	+0.04	+0.41	+0.36	+0.45	+0.40
Top 10%	+0.04	+0.04	+0.38	+0.35	+0.42	+0.39
Bottom 50%	−0.02	−0.02	−0.17	−0.16	−0.18	−0.18
Bottom 10%	−0.00	−0.00	−0.02	−0.02	−0.02	−0.02
<i>Panel B: Tax increases</i>						
	OBRA 1993		ATRA 2013		Combined	
Δ	2002	2019	2002	2019	2002	2019
<i>Growth (%)</i>						
Output	−0.19	−0.51	0.00	−0.26	−0.19	−0.77
Wage	−0.15	−0.44	0.00	−0.24	−0.15	−0.67
<i>Inequality measures</i>						
Wealth Gini (pp)	−0.04	−0.10	0.00	−0.05	−0.04	−0.15
Top 10%/median (%)	−0.25	−0.68	0.00	−0.32	−0.25	−1.00
Wealth poverty (%)	−0.74	−1.67	0.00	−0.82	−0.74	−2.37
<i>Income share shift (pp)</i>						
Top 1%	−0.14	−0.16	0.00	−0.33	−0.14	−0.48
Top 10%	−0.14	−0.16	0.00	−0.31	−0.14	−0.47
Bottom 50%	+0.06	+0.07	0.00	+0.14	+0.06	+0.21
Bottom 10%	+0.01	+0.01	0.00	+0.02	+0.01	+0.03

Notes: Reform-only counterfactuals: each CF subtracts only the named reform’s reform-year τ_K shock from the factual path going forward, leaving all other reforms intact. Reform-year shocks: Reagan -0.005 , Bush -0.040 , OBRA $+0.020$, ATRA $+0.032$. Growth and “Top 10%/median” / “Wealth poverty” rows: (factual $-$ counterfactual) / counterfactual $\times 100$. “Wealth Gini” row: percentage-point shift in Gini coefficient $\times 100$. “Wealth poverty” is the fraction of households with wealth below half the median. Income share shifts in percentage points. “Combined” stacks the panel’s two reform shocks additively. ATRA 2013 entries at 2002 are zero because the reform has not yet occurred.

5.2.1 Household parameter distributions and sorting at the cross-section

The macro tax counterfactuals just reported take (z_i, e_i, β_i) recovered from PSID and SCF as policy-invariant primitives. We now turn to those primitives directly: their lifecycle profiles, three-decade trends, variance composition, and age-conditional sorting. Section 5.2.2 then returns to the same data to test whether the structural parameters actually shift around the reforms studied in Section 5.1.4.

Lifecycle profiles. The cross-section reveals a clean lifecycle pattern in β that the macro aggregate masks. Figure 8 reports survey-weighted *medians* of (z_i, e_i, β_i) by 5-year age bin, pooling all waves within each survey. We use the median, not the mean, because all three recovered parameters are strongly right-skewed (Figure L.17 in Appendix L.3), so the weighted mean is pulled up by a thin upper tail. For productivity the choice also matters for selection: the wage sample conditions on workers, and because high earners are the ones who stay employed past age 60, the mean rises into old age while the median does not.

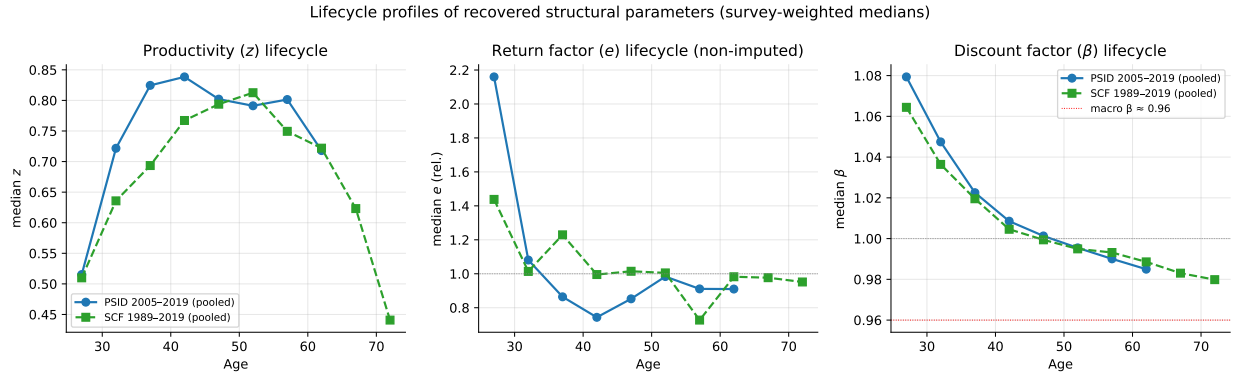


Figure 8: Lifecycle profiles of (z_i, e_i, β_i) , pooled across waves

Notes: Survey-weighted medians by 5-year age bin, pooling all waves within each survey (PSID 2005–2019; SCF 1989–2019). The e panel uses non-imputed observations only and is display-normalized by each survey’s pooled median, since e has no meaningful life cycle and the surveys differ only in level. PSID ends at the 60–64 bin: the processed PSID sample conditions on a positive observed wage and is capped at head age 65, so retired heads, for whom neither the wage nor the labor margin identifies the parameters, are absent. Distributions and a within-person (fixed-effect) version of the z panel are in Appendix L.3 (Figures L.17 and L.16).

Three patterns are robust across both surveys.

Productivity z_i traces a Mincer hump. Median z rises from ~ 0.5 at age 27 to a peak near age 42 (PSID, 0.84) or 52 (SCF, 0.81), then declines smoothly, reaching 0.44 in SCF by age 72. The two surveys track each other closely over the shared age range. The hump is a property of the median: the weighted mean instead climbs into old age, an artifact of right-skew and of high earners selecting into work past 60. A within-person profile that

removes each person’s fixed effect (76.9% of $\log z$ variance, Table 7) shows the same shape, peaking around age 32–52 and declining after, and so confirms that the hump is genuine aging rather than composition (Figure I.16).

The return factor e_i has no life cycle. The weighted median of e is flat across age in both surveys. The apparent age variation in the mean (not shown) is a heavy-tail artifact of the capital-income-to-wealth ratio, which a few low-wealth households with reported capital income drive; the median removes it.

Discount factor β_i declines with age, and the two surveys agree. The weighted median β falls from 1.08 (PSID) and 1.06 (SCF) at age 27 to about 0.98–0.99 by age 62, converging on the macro level ~ 0.96 from above. Because both surveys are now identified the same way, from the static labor FOC, the two profiles nearly coincide across the whole life cycle (within 0.02). Young households appear to “save more than the macro return predicts” ($\beta > 1$): they are accumulating wealth faster than the macro r alone would explain. This is consistent with housing appreciation, family transfers, and front-loaded retirement contributions, all of which the model’s saving identity attributes to β_i . Late-life β_i converges to the macro level from above, consistent with mature households whose wealth dynamics are governed primarily by the model’s gross return.

Wealth a_i rises monotonically with age in both surveys. Median wealth grows tenfold across the life cycle in PSID and 80-fold in SCF (whose right tail captures retirement-age accumulation that PSID misses).

Distributional dynamics over three decades. The 10 SCF and 8 PSID waves also let us trace how the cross-section of recovered parameters evolves over nearly three decades: the dispersion of β_i widens, its wealth gradient stays large and stable, and productivity–wealth sorting strengthens in PSID while weakening at the SCF top. We report these dynamics, with Figure I.15, in Appendix I.3.

Variance decomposition: fixed effect, lifecycle, and noise. The PSID panel sharpens the question of where each parameter’s variation comes from. Of the total cross-sectional variance, how much is a permanent individual fixed effect, how much is a within-person life-cycle trend, how much comoves with the wave (aggregate), and how much is idiosyncratic noise? Table 7 reports the decomposition.

The verdicts contrast sharply. *For $\log z$, the fixed effect dominates.* Person FE absorbs 77% of variance; idiosyncratic shocks 22%; the within-person life-cycle trend only 0.3%. Productivity is overwhelmingly a permanent type, with within-person AR(1) of 0.77 biennial (0.88 annualized, close to the macro calibration $\rho_z = 0.95$). The MRCE’s AR(1) z process

Table 7: Variance decomposition of recovered parameters (PSID panel, 6,943 persons / 30,241 obs)

Parameter	Total var.	Person FE	Age (lifecycle)	Wave (agg.)	Idiosyncratic
$\log z$	0.680	77.0%	0.3%	0.3%	22.4%
β (static, baseline)	0.045	47.1%	0.1%	0.0%	52.8%
β (saving, dynamic)	3.297	29.6%	0.0%	0.1%	70.2%

Notes: Each row decomposes the total panel variance into a person fixed effect (variance of person-mean), a within-person age trend (variance explained by a third-order polynomial in age after demeaning within person), a wave/aggregate component (variance of within-person residual averaged by wave), and an idiosyncratic residual. Restricted to PSID persons observed in ≥ 2 waves. The baseline static-FOC β has both far smaller total variance and a larger permanent share than the dynamic saving-equation β (last row), whose wide dispersion is mostly transitory wealth-shock noise; the saving β is shown unclipped.

matches the within-person idiosyncratic component well (its calibrated unconditional sd ~ 0.39 matches the within-person sd $\approx 0.83\sqrt{0.22} \approx 0.39$), but the data also reveal a much wider permanent component (sd $\approx 0.83\sqrt{0.77} \approx 0.73$) absent from the macro calibration. The cross-sectional Mincer hump in z by age (Figure 8) reflects partly within-person life-cycle accumulation but mostly between-person composition: different age groups draw from different fixed-effect distributions.

For β , idiosyncratic noise dominates. Person FE absorbs only 30% of variance; idiosyncratic shocks 70%; the within-person life-cycle trend essentially zero ($< 0.1\%$). Within-person biennial AR(1) is -0.12 (annualized -0.34): individuals do not have stable β paths, only mildly mean-reverting noise around their personal average. The lifecycle decline of β visible in the cross-sectional age profiles is therefore mostly a between-person composition effect, not within-person aging: at older ages the sample shifts toward households whose fixed-effect component happens to be lower, not toward individuals who became less patient. The fixed-effect component (the 30%) reflects cross-sectional preference heterogeneity; the dominant idiosyncratic component reflects realized wealth shocks (housing appreciation, business income volatility, family transfers) that the saving identity attributes to β and that mean-revert at the individual level.

The wave (aggregate) component is tiny for both. Less than 0.3% of $\log z$ or β variance comoves with the wave. This is direct evidence that the macro $Z_{L,t}$ and β_t time series do not pass through the household-level inversion. The aggregate wedges are economy-wide objects (utilization, expectations) that a household-level decomposition does not register, consistent with the discussion in Section 5.3.

Sorting peaks in prime working ages. Table I.8 (Appendix I.3) reports cross-parameter correlations within age bins, pooled across waves. In SCF, $\rho(\beta_i, z_i)$ is weak at age 25–29 (-0.18), strengthens to -0.34 at age 40–49, and weakens again at retirement age (-0.07 at 65–74). The wealth gradient $\rho(\beta_i, \log a_i)$ follows a similar inverted-U: -0.41 at 25–29, peaks at -0.54 at age 40–49, and weakens to -0.10 at age 65–74. Under the static baseline PSID shows the same inverted-U: $\rho(\beta_i, z_i)$ strengthens from -0.14 at age 25–29 to -0.20 to -0.26 in prime ages, and $\rho(\beta_i, \log a_i)$ peaks at -0.40 at age 40–49. (Under the dynamic route the PSID correlations are near zero in every bin, because the saving-equation noise dilutes the gradients; Table 7.) The two surveys thus agree on the sorting structure as well as the level. The sorting structure that drives the macro aggregate is strongest in prime working ages, when productivity differences manifest in wages, wealth differentials are largest, and life-cycle effects accumulate. Outside this window, sorting is weaker: at young ages most households are in early-career wealth-poor positions, while at retirement age wealth is decoupled from current labor productivity.

5.2.2 Policy invariance: a Lucas-critique test

The tax-reform counterfactuals of Section 5.1.4 treat the structural parameters (z_i, e_i, β_i) as policy-invariant: the same households face the counterfactual tax sequence and respond through the closed-form MRCE policy at the new prices. The household-level inversion provides a direct test of that assumption around the same four reforms.

Policy invariance: parameters stable around major tax reforms. The variance decomposition showed less than 0.3% of within-person z or β variance comoves with the wave. A direct policy test reinforces the same conclusion. The 27-year SCF window and 14-year PSID window contain four major federal tax reforms split between cuts and increases: the Omnibus Budget Reconciliation Act of 1993 (top rate 31% $\rightarrow$ 39.6%, an increase, between SCF 1992 and 1995); the Economic Growth and Tax Relief Reconciliation Act 2001 (a cut, between SCF 2001 and 2004); the American Taxpayer Relief Act 2013 (top rate 35% $\rightarrow$ 39.6% and capital-gains rate 15% $\rightarrow$ 20%, an increase, between SCF 2010–2013 and PSID 2011–2013); and the Tax Cuts and Jobs Act 2017 (corporate rate 35% $\rightarrow$ 21%, between SCF 2016–2019 and PSID 2017–2019). Table 8 reports the recovered moments in the wave just before and just after each reform.

The cross-sectional distributions are essentially unchanged across all six pre/post comparisons. The median β moves by at most 0.02 in either survey; the standard deviation of $\log z$ is unaffected; the wealth gradient $\rho(\beta_i, \log a_i)$ stays in its narrow band. Differences are not systematically signed by the direction of the policy change.

Table 8: Parameter moments around major US tax reforms

Reform	Source	Pre–Post	β_{med}	β_{cap}	$\text{sd}(\log z)$	$\rho(\beta, \log a)$
<i>Tax increases</i>						
OBRA 1993	SCF	1992–95	1.00 $\rightarrow$ 1.00	0.92 $\rightarrow$ 0.76	0.89 $\rightarrow$ 0.88	–0.55 $\rightarrow$ –0.50
ATRA 2013	SCF	2010–13	1.01 $\rightarrow$ 1.01	0.88 $\rightarrow$ 0.88	0.93 $\rightarrow$ 0.91	–0.42 $\rightarrow$ –0.42
ATRA 2013	PSID	2011–13	1.00 $\rightarrow$ 1.01	0.94 $\rightarrow$ 0.92	0.86 $\rightarrow$ 0.86	–0.30 $\rightarrow$ –0.35
<i>Tax cuts</i>						
EGTRRA 2001	SCF	2001–04	1.00 $\rightarrow$ 1.00	0.92 $\rightarrow$ 0.89	0.89 $\rightarrow$ 0.90	–0.55 $\rightarrow$ –0.48
TCJA 2017	SCF	2016–19	1.01 $\rightarrow$ 1.01	0.85 $\rightarrow$ 0.88	0.94 $\rightarrow$ 0.91	–0.47 $\rightarrow$ –0.46
TCJA 2017	PSID	2017–19	1.01 $\rightarrow$ 1.02	0.96 $\rightarrow$ 0.95	0.86 $\rightarrow$ 0.82	–0.33 $\rightarrow$ –0.32

Notes: Pre/post moments are computed within the consistent estimation sample for each survey, with population weights. The 1995 SCF β_{cap} outlier (0.76) reflects a wave-specific composition shift, not a sustained policy response: β_{cap} recovers to 0.91 by 1998. Under the static baseline PSID and SCF move by comparable, negligible amounts.

The aggregate moments above could still mask offsetting movements across households, so the decisive test is micro-level and uses PSID’s panel, comparing the *same* household’s β_i before and after each PSID-spanning reform. Two checks go beyond the aggregate. First, the entire *distribution* of within-person $\Delta\beta_i$ around reforms is indistinguishable from normal times: across the two reform-spanning wave pairs (ATRA 2013, TCJA 2017) the mean is 0.00 and the standard deviation 0.24, against mean –0.00 and standard deviation 0.23 in the five non-reform pairs (Kolmogorov–Smirnov $D = 0.02$), so reforms do not shift or even widen the spread of individual changes. Second, because a symmetric distribution could still hide a response concentrated in the affected households, we run a triple-difference on the group most exposed to the capital-tax reforms, the top wealth quartile measured pre-reform. Top-wealth households do drift β upward by +0.015 relative to the rest, but they do so in *every* period (the same +0.016 in non-reform pairs, a mean reversion from their low β), and the reform adds nothing on top: the differential response specific to the reform is –0.001 ($t = -0.1$). Recovered preferences do not move around major tax reforms in the aggregate, in the distribution of individual changes, or among the households the reforms hit hardest.

This is a Lucas-critique-style test that the structural identification passes. The macro inversion’s tax counterfactuals (Section 5.1.4) treat structural parameters as policy-invariant, and the micro panel directly verifies that they are. Because the tested events span both increases and cuts, the policy-invariance result is not specific to the direction of any single political episode.

Frequency scope of the invariance. This invariance is established at the annual frequency of the structural inversion, and against policies that act through contemporaneous real incentives, the federal tax reforms. It should be read as invariance at the business-cycle

frequency our applications use, not at all frequencies. Policies that work through the expectations channel, in particular monetary forward guidance and central-bank communication, act over horizons of weeks and months and may move household saving only transiently; a response of that kind would be averaged out under annual aggregation, so the present test is silent on it. Because the MRCE loads the forward-looking content onto β (Section 3.4), a higher-frequency inversion, fed by higher-frequency data on both household balance sheets and identified policy shocks, could plausibly detect a β response to such expectations-channel policy that the annual analysis cannot. We therefore read the recovered β as structural at the annual, business-cycle frequency relevant here, and leave its invariance to higher-frequency, expectations-channel policy as an open question for higher-frequency data.

5.3 Macro and micro β as two scales

Macro–micro parallel and contrast. The paper recovers two discount factors, the macro β_t from aggregate data and the micro β_i distribution from PSID and SCF, and likewise two versions of labor productivity, the aggregate $Z_{L,t}$ and the individual z_i . Each pair plays a related but different role. The aggregate $Z_{L,t}$ is the time-varying common component of effective labor productivity (utilization, composition, matching), moving at business-cycle frequencies; individual z_i is the cross-sectional permanent characteristic. The aggregate β_t is the time-varying expectations / saving-margin wedge; individual β_i is the cross-sectional preference characteristic plus transitory wealth-shock noise. The macro inversion identifies the aggregate (time-series) component of each parameter; the micro inversion identifies the individual (cross-sectional) component. They are conceptually orthogonal: aggregate cyclical dynamics and cross-sectional heterogeneity. The framework’s joint identification of both, via the same closed-form policies inverted at two scales, is the bridge between them.

Three threads pull the macro and micro inversions together.

Cross-survey replication is the strongest empirical content. PSID and SCF apply the same static identification to disjoint samples spanning three decades, and a second, dynamic identification is available on PSID alone. The two surveys recover a median household β within two percentage points of each other and signed-equal cross-sectional gradients in every wave; their capital-weighted aggregates differ only by how deep each survey reaches into the wealthy top. On PSID, the dynamic (saving-equation) and static (labor-FOC) routes use unrelated data moments yet recover the same median β (Appendix I.3), so the result is robust to the identification as well as to the sample. The MRCE structure (closed-form policies, time-invariant idiosyncratic processes, two-orthogonal-shocks decomposition) survives

this replication. Two independent datasets agreeing, and two unrelated identification routes within PSID agreeing, is the strongest evidence the framework’s micro side can produce.

Macro and micro β measure different economic objects. The macro β_t describes growth in universe productive capital (BEA fixed assets summed across all sectors); the micro β_i describes household balance-sheet dynamics. The gap between the survey $\hat{\beta}_{\text{cap}}$ and the macro $\beta_t = 0.967$ is dominated by the wealthy quartile, and it differs across surveys by how deep each reaches into that quartile: SCF, whose list sample captures the ultra-wealthy, sits ~ 8 pp below the macro level ($\hat{\beta}_{\text{cap}} = 0.877$), while PSID, which misses the very top, sits only ~ 2 pp below (0.936). Median Q4 wealth shrinks at $-1.9\%/yr$ in PSID, while universe-level capital grows at $+1\%/yr$ in 2005–2019. The wedge is the household-vs-universe accumulation gap, not measurement error.³

Macro β as an expectations channel; cross-sectional β as preference heterogeneity. Section 4 reports a strong negative correlation $\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.858$ in macro first differences: aggregate patience rises sharply when labor productivity falls. The micro decomposition shows that this correlation does not arise at the household level. Cross-sectional $\rho(\beta_i, z_i)$ is small in any wave (below -0.05 in PSID, between -0.11 and -0.44 in SCF). The time series of micro $\hat{\beta}_{\text{cap},t}$ co-moves only weakly with $Z_{L,t}$ at the wave frequency. Within-person regressions of $\Delta \beta_i$ on $\Delta Z_{L,t}$ are insignificant. Individual households do not literally become more patient when aggregate labor productivity falls. A natural reading is that the macro β_t captures aggregate *expectations* that co-move with $Z_{L,t}$: when labor productivity deteriorates, agents revise downward their expected paths of future income and returns, and aggregate saving rises in response, alongside corporate cash hoarding and other economy-wide forces that the household-level inversion does not register. The cross-sectional β_i heterogeneity, by contrast, identifies preference differences, and the within-person evidence above shows it has both a persistent (preference) component and a transitory (wealth-shock) component. The two objects, aggregate expectations and cross-sectional preferences, are conceptually distinct, and the micro inversion separates them.

Scope of the tax counterfactuals. The expectations reading of β above has a direct consequence for Section 5.1.4. The counterfactuals treat (z_i, e_i, β_i) as policy-invariant primi-

³One might suspect housing wealth, with its asset-price volatility, accounts for the gap. The opposite holds: re-running the PSID inversion using PSID’s financial wealth alone (excluding housing equity) yields $\hat{\beta}_{\text{cap}}^{\text{fin}} = 0.932$, a -2.5 pp gap to the macro level, marginally wider than the -2.2 pp gap from total net worth. The wedge is not driven by housing per se but by the broader mismatch between household balance-sheet dynamics and universe-level productive-capital growth.

tives, so any channel through which a new policy reshapes agents’ expectations, and thereby their inferred patience, is by construction absent from the simulated paths. The Lucas-style test of Section 5.2.2 offers empirical comfort that the four reforms in our sample windows did not detectably move β_i , but the four episodes were moderate in scale and largely anticipated; the test does not speak to truly novel or unanticipated regimes for which an expectations response is plausible. The reported magnitudes are therefore best read as the sorting-channel response under the maintained policy-invariance hypothesis, with any expectations-induced movement in β_i left as a separate channel that would temper or amplify them depending on the sign of agents’ belief revision.

6 Conclusion

This paper develops a heterogeneous-household model that can be inverted on data rather than solved forward. A preference structure with closed-form, forecast-invariant policies collapses the wealth distribution to three moments, aggregate capital and two sorting covariances, and makes the equilibrium independent of household expectations. The same policies can then be read backwards at two scales. From U.S. aggregates we recover productivity, patience, and labor efficiency period by period, and from PSID and SCF we recover the discount factor, ability, and return type of individual households. No detrending or stationarity assumption is needed: trend, cycle, and distributional change are explained within one structure.

The application gives several results. On the aggregate side, the postwar cycle has two dimensions, with TFP nearly orthogonal to a block of labor efficiency and patience; wealth concentration is the endogenous result of return-wealth sorting under TFP growth; the model forecasts output and labor better than a recursive AR benchmark; and persistent capital-tax reforms, in either direction, produce distributional drift that lasts for decades. On the household side, the same closed-form identification applied to two independent surveys recovers a median household discount factor that agrees within 2 percentage points; their capital-weighted aggregates differ by more and each sits below the macro estimate, by margins that track how far each survey’s wealth coverage reaches into the low- β top. No household’s recovered preferences move around four major tax reforms. The cross-survey agreement on the typical household, together with the stability across reforms, is the strongest evidence the framework’s micro side offers.

Two points deserve emphasis. First, the reduction is exact for contemporaneous allocations and for the aggregate capital law of motion, but the forward evolution of the two covariances is not closed in three moments, so the cross-section is still simulated. The gain

over standard methods is that the simulation uses exact policies and an exact capital recursion rather than an estimated forecasting rule. Second, the recovered discount factor is semi-structural: forecast invariance removes the expectational fixed point but bundles patience and expectation into one number, so neither β_t nor β_i is pure preference (Section 5.3 separates the two by scale). This bounds the policy-invariance result: the reforms we test were moderate and largely anticipated, so a genuinely new regime that shifts expectations could move the wedge component even if the preference component is stable, and the stability we document is then stability of the composite. Disentangling the components, which requires bringing observed consumption into the household inversion as a second equation, is left to future work.

More broadly, the framework suggests inversion-based structural macroeconomics as a practical alternative to forward solving: data go in, structural shocks come out, and forecasts and counterfactuals follow at once, with re-inversion taking seconds when new data arrive. Three extensions are natural. Embedding the MRCE in a New Keynesian model would let us study how monetary policy interacts with distributional dynamics. Moving to quarterly data would bring the forecasting closer to real-time use. Applying the same policies across countries, where harmonized household surveys exist, would measure how institutions and culture shape patience, a deep parameter that forward-solving methods usually calibrate rather than recover.

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